MIL-W-52125B
19 July 1968

SUPERSEDING
MIL-W-52125A(MO)
29 July 1964

MILITARY SPECIFICATION

WASHING AND SCREENING UNIT, AGGREGATE, ELECTRIC-MOTOR-DRIVEN, 75-TON-PER-HOUR CAPACITY

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a semitrailer-mounted, pneumatic-tired, electric-motor-driven aggregate washing and screening unit with dolly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

W-C-586	- Conduit Outlets and Entrance Caps, Electrical; Cast Metal - for Shore Use.
FF-B-171	- Bearings, Ball, Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187	- Bearing, Roller, Tapered.
QQ-S-781	- Steel, Strapping, Flat.
QQ-S-790	- Steel, Strapping, Round (Bare and Zinc-Coated).
SS-R-406	- Road and Paving Materials; Methods of Sampling and Testing.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

FSC 3820

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TT-V-121	- Varnish, Spar, Water-Resisting.
UU-T-81	- Tags, Shipping and Stock.
WW-C-566	- Conduit, Metal, Flexible.
WW-C-581	- Conduit, Metal, Rigid; and Coupling, Elbow, and Nipple, Electrical Conduit; Zinc-Coated.
WW-V-58	- Valves, Gate, Cast-Iron; 125- and 250-Pound, Screwed and Flanged (for Land Use).
ZZ-H-561	- Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water.
GGG-J-51	- Jacks, Mechanical and Hydraulic, Hand Operated.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-C-650	- Crates, Wood, Open and Covered.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof for Packaging.

Military

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-T-740	- Trucks, 5 Ton, 6 x 6, Military Design, M39 Series.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-C-3432	- Cable and Wire, Electrical (Power and Control; Semi-Flexible, Flexible and Extra Flexible, 300 and 600 Volts).
MIL-F-3541	- Fittings, Lubrication.
MIL-C-3774	- Crates, Wood; Open, 12,000 and 16,000 Pound Capacity.
MIL-G-3859	- Grease Guns, Hand, High Pressure, Lever-Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).

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MIL-H-3992	- Hose and Hose Assembly, Rubber; Air and Vacuum Brake, Automotive.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Aluminum, and Aluminum Alloys.
MIL-S-8660	- Silicone Compound.
MIL-P-10360	- Pumps, Centrifugal, for Water.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-R-11461	- Rod, Ground, (With Attachments).
MIL-T-12459	- Tire, Pneumatic; for Military Ground Vehicles.
MIL-S-12514	- Starters, Motor; Across-the-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100-Hp, Manual and Electric.
MIL-C-13486	- Cable, Special Purpose, Electrical; Low-Tension, Heavy-Duty, Single-Conductor and Multi-Conductor.
MIL-P-13983	- Paint, Temporary, Lusterless, Gasoline Removable.
MIL-I-15126	- Insulation Tape, Electrical, Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive.
MIL-H-15424	- Hand Tools; Packaging of.
MIL-V-16721	- Valves, Foot, Low Pressure (for Water).
MIL-L-21260	- Lubricating Oil, Internal-Combustion Engine, Preservative.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-N-25027	- Nut, Self-Locking, 250°F., 450°F., and 800°F., 125 KSI FTU, 60 KSI FTU, and 30 KSI FTU.
MIL-P-46093	- Primer Coating, Synthetic (for Brake Drums).
MIL-T-62021	- Truck, 10 Ton, 6 x 6, M121 A1 Series.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-15	- Graphic Symbols for Electrical and Electronic Symbols.
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-421	- Chain, Roller; Power Transmission and Conveyor, Flat Link Plates, Single Pitch, Single and Multiple Strand: Connecting Links and Attachment Links.
MIL-STD-423	- Chain, Roller; Conveyor, Standard and Large Rollers, Flat Link Plates, Double Pitch, Single Strand; Connecting Links and Attachment Links.
MIL-STD-424	- Chain, Roller; Power Transmission, Offset Sidebars, Single-Pitch.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-619	- Unified Soil Classification System for Roads, Airfields, Embankments and Foundations.
MIL-STD-642	- Identification Marking of Administrative, Combat, and Tactical Transport Vehicles.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1224	- Visual Inspection Guide for Pneumatic Tires (Nonaircraft).
MIL-STD-1261	- Welding Procedures for Constructional Steels.
MS21322	- Clamps, Loop - Aluminum - Plain and Cushioned.
MS21333	- Clamps, Loop - Steel - Plain and Cushioned.
MS21334	- Clamps, Loop - Double, Cable or Tube, Steel, Plain and Cushioned.
MS27142	- Connector, Plug, Electrical - Pin Contact, No. 12, 14 & 16 A.W.G., Waterproof.
MS27143	- Connector, Plug, Electrical - Socket Contact, No. 12 A.W.G., Waterproof.
MS27144	- Connector, Plug, Electrical: Socket Contact, No. 14 & 16 A.W.G., Waterproof.

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MS27145	- Connector, Plug, Electrical: No. 14 A.W.G., Double Contact, Waterproof.
MS27146	- Connector, Receptacle, Electrical - No. 12, 14 & 16 A.W.G., Double Contact, Waterproof.
MS27147	- Adapter, Connector - 3-Way No. 12, 14 & 16 A.W.G. Waterproof.
MS27148	- Contact; Pin, Electrical Connector: No. 12, 14 & 16 A.W.G.
MS33800	- Connector, Receptacle, Electrical - Pin Contact, No. 12, 14 & 16 A.W.G., Waterproof.
MS35387	- Reflector, Indicating, Clearance.
MS35390	- Tire, Pneumatic: Low Platform Trailer, Earthmover, Grader, and Amphibious Vehicle.
MS35392	- Inner Tube: Pneumatic Tire.
MS35423	- Light, Marker, Clearance - Service.
MS35424	- Light, Marker, Clearance - Blackout.
MS35746	- Coupling, Automotive - Air Brake Lines.
MS35829	- Hinge, Continuous (Piano), .075 Thickness, .187 and 250 Pin Dia.
MS35844	- Plug, Machine Thread; Magnetic: Drain.
MS39020	- Band, Marker, Blank.
MS39062	- Circuit Breakers: Thermal, 30 Volt DC, Automatic, Waterproof, 15 to 30 Amp.
MS39127	- Tray Screen Sizes and Edging for Aggregate Screens.
MS49006	- Plugs, Pipe, Magnetic - Headless; Aluminum, Iron or Steel.
MS51118	- Pintle Assembly, Towing - 40,000 Lbs Capacity, Manual Release.
MS51330	- Stop Light - Taillight, Vehicular - 24 Volt, Blackout Tail, Blackout Stop.
MS51337	- Coupler, Drawbar, Ring - Heavy Duty.
MS53004	- Valve, Relay, Emergency-Breakaway, Air Brake.
MS53007	- Plate, Identification - Emergency and Service.
MS35034	- Fifth Wheel Assembly - 36-Inch Universal, Truck-Tractor, 45,000 Lbs. Capacity.
MS53038	- Kingpin, Fifth Wheel - Universal.
MS53039	- Socket, Kingpin, Fifth Wheel - Universal.

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MS53040	- Retainer, Kingpin, Fifth Wheel - Universal.
MS75020	- Connector, Plug, Electrical -12 Contact, Intervehicular 24 Volt, Waterproof.
MS75021	- Connector, Receptacle, Electrical - 12 Contact Intervehicular, 24 Volt, Waterproof.
MS500008	- Truck, Tractor, 5 Ton, 6 x 6, M52.
MS500010	- Truck, Dump, 5 Ton, 6 x 6, M51.
MS500070	- Truck, Tractor: 10 Ton, 6 x 6, 181.5 Inch Wheelbase, M123A1C.
MS500081	- Dolly, Trailer Converter, 15 Ton, 4 Wheel Military Design, M354.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

AB-1 - Circuit Breakers, Molded Case.
IC-1 - Control, Industrial.
MG-1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, N. Y. 10017).

FORGING INDUSTRY ASSOCIATION

Forging Industry Handbook

(Application for copies should be addressed to the Forging Industry Association, Illuminating Bld., Cleveland, Ohio 44101).

RUBBER MANUFACTURERS ASSOCIATION, INC.

Standard Specifications for Drives Using Narrow V-Belts (3V, 5V, and 8V Cross Sections).

(Application for copies should be addressed to the Rubber Manufacturers Association, Inc., 444 Madison Avenue, New York, N. Y. 10022).

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UNITED STATES OF AMERICA STANDARDS INSTITUTE

B6.6 - Gear Tolerances and Inspection.
B55.1 - Specifications for Multiple V-Belt Drives.
Z35.1 - Industrial Accident Prevention Signs, Specification for.

(Applications for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, N. Y. 10016).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A242 - High-Strength, Low-Alloy Structural Steel.
A536 - Ductile Iron Castings.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd., Chicago, Illinois 60606.)

GENERAL SERVICES ADMINISTRATION

Code of Federal Regulations
Title 23 - Highways and Vehicles

(Application for copies should be made to the Superintendent of Documents, Experiment Printing Office, Washington, D.C. 20402.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

Specification E-3-1961 - Electrical Lighting and Wiring for Commercial Vehicle Bodies.
National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street NW, Washington, D.C. 20036).

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TIRE AND RIM ASSOCIATION, INC.

Tire and Rim Association Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc., Comand Building, 34 North Hawkins Avenue, Akron, Ohio 44313.)

DEPARTMENT OF TRANSPORTATION

Motor Carrier Safety Regulations

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

NATIONAL SAND AND GRAVEL ASSOCIATION

Circular No. 80, April 1959

(Application for copies should be addressed to the National Sand and Gravel Association, 527 Munsey Building, Washington, D.C. 20004.)

TRUCK-TRAILER MANUFACTURERS ASSOCIATION, INC.

RP3-1-l2 Trailer Air Brake System Test
RP4-1-4 Truck Trailer Supports
RP6-2-9 Mounting or Required ICC Lighting Devices

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, 1413 K Street NW, Washington, D.C. 20005.)

AMERICAN SOCIETY OF AGRICULTURAL ENGINEERS

ASAE S276.1 - Slow-Moving Vehicle Identification Emblem.

(Application for copies should be addressed to the American Society of Agricultural Engineers, P.O. Box 229, St. Joseph, Michigan 49085.)

FRICTION MATERIALS STANDARDS INSTITUTE, INC.

Automotive Data Book

(Application for copies should be addressed to the Friction Materials Standards Institute, Inc., 370 Lexington Avenue, New York, N.Y. 10017.)

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VEHICLE EQUIPMENT SAFETY COMMISSION

Regulation V-3

(Application for copies should be addressed to the Vehicle Equipment Safety Commission, 1026 17th Street NW, Washington, D.C. 20036.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, N.Y. 10001.)

NATIONAL BUREAU OF STANDARDS

Handbook H-28 - Screw-Thread Standards for Federal Services.

(Application for copies should be obtained from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Description. The washing and screening unit (hereinafter called "unit") shall consist of a pneumatic-tired semitrailer, with dolly, on which is mounted an electric-motor-driven revolving scrubber, triple-deck vibrating screen, single-screw-type sand dehydrator, chutes, plumbing, components, and accessories. Provisions shall be made for feeding by a portable conveyor into a scrubber receiving hopper when in operation. Waterpipe, fixtures and flume shall be provided to control the water required for washing the aggregate.

3.1.1 Capacity. The unit shall process a minimum of 75 tons per hour (TPH) of 1-1/2-inch and smaller rock aggregate produced in quarry or gravel pit operations containing not more than 2 percent of material larger than 1-1/2-inch and not more than 40 percent fines passing a 1/4-inch screen, and shall contain not less than 8 percent CL soil as defined in MIL-STD-619. The material shall have a moisture content above the plastic limit.

3.1.2 Physical characteristics. The physical characteristics of the unit with dolly in operational condition for processing rock, all components, attachments, accessories, equipment, and tools shall be as specified. All measurements shall be made with the unit on level ground, and the inflation pressure of the tires shall be as recommended by the tire manufacturer for operation at 30 mph highway speeds. Axle loads shall be measured at the ground under the tires.

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Established component and vehicular rating and capacities shall not be increased to meet the requirements of this specification. The dolly drawbar shall be in the straight-ahead towing position. The minimum practicable number of components may be removed to meet the reducible height specified in (f). Any sectionalization and reassembly required shall each be accomplished in not more than 3 hours without cutting or welding.

(a)	Total weight	60,000 lbs., max.
(b)	Fifth wheel load limit	30,000 lbs., max.
(c)	Axle load limit, each axle	18,000 lbs., max.
(d)	Overall length	600 in., max.
(e)	Overall width	120 in., max.
(f)	Reducible highway transport height	162 in., max.
(g)	Ground clearance (SAE J894)	9.5 in., min.
(h)	Ground clearance excluding axles with dolly and with either truck-tractor (M123 and M52)	12 in., min.
(i)	Upper fifth-wheel height	
	(1) Semitrailer level	58 in.
	(2) for 10 ton M123 (see MS500070)	64 in.
	(3) for 5 ton M52 (see MS500008)	55 in.
(j)	Landing gear (front) clearance radius ("LWC" of SAE J701)	90 in., min.
(k)	Swing radius ("SR" of SAE J701)	58 in., max.
(l)	Kingpin setting range ("KP" of SAE J701)	21 in. through 36 in.
(m)	Center-of-gravity height	110 in., max.

3.1.3 Ease of Maintenance. All major assemblies and installed attachments shall be accessible for maintenance and repair utilizing common tools and special tools furnished with the unit. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage and with minimum disturbance to other components of the unit. Ease of maintenance provisions shall incorporate to the maximum practicable extent features insuring operating clearances and facilitating maintenance and servicing operations. Unless otherwise specified herein, covers, plates, panels, and guards which must be removed for adjustment, maintenance, or component removal shall be held in place by nonremovable studs, washers, lock-washers, and nuts. Where practicable, access shall be provided by a hinged door, and hinges shall be welded to door and frame of opening. Such doors shall be provided with a hinged or nonhinged hasp which mates with a nonremovable stud, and be held closed by a lockwasher and nut.

3.2 Preproduction model. The supplier shall furnish a unit for examination and test within the time frame specified (see 6.2) to prove, prior to starting production, that his production methods and choice of design detail will produce units that comply with the

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requirements of this specification (see 6.3). Examination and tests shall be those specified herein.

3.3 Initial production. When specified (see 6.2), the supplier shall furnish to the Government one or more units (see 6.2) for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. All material shall be new and unused. Material not specified shall be selected by the supplier as being compatible with the intended use in accordance with the engineering standards of the earthmoving and construction equipment industry, and as set forth in the publications of the American Society for Testing and Materials (ASTM), Society of Automotive Engineers including SAE Handbook (SAE), American Society of Agricultural Engineers (ASAE), and United States of America Standards Institute (USA STD). Unless otherwise specified herein, rubber products furnished shall not exceed 8 calendar quarters in age from date of cure of the rubber to date of acceptance of the unit by the Government. Threaded parts shall be in accordance with SAE Standards and Handbook H28. Maximum practical use shall be made of interchangeable hardware and fastening devices and of a minimum number of types and sizes of studs, bolts, screws, washers, nuts, and similar common (standard) parts.

3.5 Stress limitations. The unit shall be constructed so that the maximum stress imposed on any structural component shall not exceed 60 percent of the yield point.

3.6 Safety. The safety characteristics of the unit shall meet the following requirements:

(a) Eliminate concealed danger created by the supplier's design.
(b) Eliminate hazards to personnel and to property.
(c) Result in supplying material having a correct safety factor.
(d) Provide needed safety and protective devices or prominent warnings. Protective devices shall not impair the operating functions.
(e) Conform to all requirements of the ICC Motor Carrier Safety Regulations, parts 190 and 193, pertaining to large semitrailers, except for rear bumper, and as specified herein.
(f) Conform to Code of Federal Regulations, Title 23, Chapter 2, Part 255, Federal Motor Vehicle Safety Standards, except as specified herein. Theoretical calculations are acceptable in lieu of destructive testing.

For exposed components and systems which are subject to high temperatures, high pressures, electrically actuated, or inherently hazardous, the safety characteristics shall provide correct safe guarding, enclosing, and insulating features. All moving parts which are of such nature or so

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located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Step surfaces shall provide a nonskid footing. Nonfunctional sharp edges, projection points, and excessive length of fasteners shall be eliminated. An unobstructed space shall be provided on the rear of the unit for mounting of an emblem identifying slow-moving vehicles. A mounting bracket socket shall be permanently installed and shall conform to SAE J725 and ASAE S276.1. An emblem and its mounting blade shall be furnished on each unit and shall be in accordance with ASAE S276.1. For other than travel, the emblem and blade shall be stowed in the toolbox.

3.7 Hoppers and chutes. All portions of hoppers, chutes, plates, and metal surfaces subjected to abrasive wear caused by aggregate movement shall be lined or faced with replaceable, abrasion-resistant steel plates not less than 3/8 inch in thickness conforming to ASTM A242. All bolt heads subjected to the flow of aggregate shall not extend beyond surfaces exposed to the material and shall be of hardened steel.

3.7.1 Screen feed chute. A feed chute shall be mounted at the feed end of the vibrator screen to receive material from the scrubber discharge chute and discharge it evenly across the full width of the screen.

3.7.2 Screen discharge hopper and chutes.

3.7.2.1 Discharge hopper. A discharge hopper shall be located under the vibrating screen to receive all material that passes through the bottom deck and pass it into the receiving hopper of the dehydrator.

3.7.2.2 Chutes. Material chutes shall be provided for the three screen decks. The ground clearance under the chutes shall be 36 inches plus or minus 2 inches when the plant is set up for operation, level both transversely and longitudinally, and supported by the stabilizing jacks (all wheels free to turn). The chute arrangement shall provide for not less than 4-1/2 foot clearance between centerlines of the discharge end of the chutes.

3.8 Electrical system.

3.8.1 Electric motors. The electric motors shall be 1,800-rpm, 440-volt, 3-phase, 60-cycle, ball-bearing, squirrel-cage induction, continuous-rated, totally-enclosed, fan-cooled, design "C" in accordance with NEMA MG-1, integral-horsepower, class "A" insulation, temperature rise 55° C. All motors shall be provided with mounting brackets. Bearings shall be in accordance with FF-B-171, type as applicable, double-seal class, packed with grease conforming to MIL-G-23827.

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3.8.2 Controls. All controls, receptacles, and switches shall be mounted on the unit so that they are accessible to the operator when the operator is standing on the ground.

3.8.2.1 Power receptacles. Six female receptacles of the 4-pole type suitable for 20-ampere, 440-volt service shall be provided. The receptacles shall be equal to and interchangeable with Joy part No. X8060-85 and shall consist of an insert equal to and interchangeable with Joy part No. X8060-87, and shall be provided with 24-inch pigtails plus a spring loaded covered mounting shell equal to and interchangeable with Joy part No. X1697. The receptacles shall be mounted in a bank and shall be accessible to the operator from the ground level. One of the six receptacles shall be provided with a motor starter as specified in 3.8.2.5 and shall be clearly identified. A main feeder receptacle of the 4-pole male type suitable for 200-amp, 440-volt service shall be provided and shall include a spring-loaded dust cover and a safety chain. The receptacle shall be equal to and interchangeable with Joy part No. X8617-26.

3.8.2.2 Main feeder disconnect. The main feeder disconnect shall be a circuit breaker conforming to NEMA AB-1. The circuit breaker shall be a nonautomatic circuit interrupter with shunt trip device. It shall provide watertight (type IV) and dust tight (type V) construction in accordance with NEMA IC-1. The breaker shall be 3-pole, rated for 225 amperes, and actuated by an external handle. The switch handle positions shall be labeled: “OFF”, “ON”, “RESET”.

3.8.2.3 Safety cutout. Two, one-position, pushbutton, switch-type, normally-open safety cutouts shall be furnished. One safety cutout shall be located on the distribution panel. The other safety cutout shall be located so as to be operable from ground level while the operator is viewing the side of the unit opposite the side on which the distribution panel is located. Each safety cutout shall be rated at 30-ampere maximum interrupting current and shall be labeled “STOP”. The safety cutouts shall be of dusttight (type V) and watertight (type IV) construction in accordance with NEMA IC-1. One safety cutout shall be located on the distribution panel. The other safety cutout shall be located so as to be operable from ground level while viewing the side of the unit opposite the side on which the distribution panel is mounted.

3.8.2.4 Distribution panel and branch circuit protective devices. The panel shall be of dusttight (type V) and watertight (type IV) construction in accordance with NEMA IC-1. The panel shall be 3-phase, 4-wire, solid neutral, with 200-ampere, main-lug capacity. Each branch circuit shall be provided with a circuit breaker in accordance with NEMA AB-1. Single-pole

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circuit breakers will be acceptable if provided with handle tie bars for each individual polyphase circuit. Each starter shall be considered as a branch circuit, and individual circuit breakers shall be provided for all starters.

3.8.2.5 Motor starters. Each motor on the unit shall be provided with a motor starter. The motor starters shall be in accordance with MIL-S-12514, type I, class 2, style B, and shall be 3-pole, alternating current, magnetic, nonreversing, across-the-line, and rated for 440-volt, 60 cycle operation. The enclosures shall be watertight (type IV) and dusttight (type V) in accordance with NEMA IC-1 and shall be provided with start-stop pushbuttons in the covers.

3.8.3 Cable assemblies.

3.8.3.1 Power cables and reels and ground rods. The main feeder cable shall consist of two identical 100-foot lengths of cable conforming to MIL-C-3432, type CO-03HGF (3/00-3/5R)1750. Each length shall be provided with a 4-pole male connector on one end and a 4-pole female connector on the other end. The cables shall be provided with safety chains and strain collars. A safety chain shall be attached to the male connector of each cable. Each connector shall have a protective cap attached to the cable with sash chain. Each cable assembly shall be equal to and interchangeable with Joy cable assembly No. X8617-25. A manually-operated reel or reels to contain the main feeder cables shall be permanently mounted on the unit. The reel or reels shall be of steel construction, ratchet type, and provision shall be made to lock the cables on the reel or reels. Three ground rods shall be furnished and shall conform to MIL-R-11461, type II, style 2, except that the ground wire shall be 15 feet long. A 1/4-inch bronze grounding stud or bolt with nut, and two internal-external toothed washers shall be provided in the frame of the unit near the distribution panel for attaching the ground-wire clamp to the unit. The ground-rod assemblies, clamps, and wires shall be stowed in the toolbox.

3.8.3.1.1 Power cable adaptor assembly. A power cable adaptor assembly shall be provided. The assembly shall consist or a 4-pole female connector on one end or a 5-foot length of cable conforming to MIL-C-3432, type CO-03HGF (3/00-3/5R)1750. A strain collar shall be provided to accommodate the safety chain on the male connector or the main feeder cable. A protective cap attached to the adaptor with sash chain shall be provided to prevent loss. The other end of the feeder cable shall have 12 inches of the cable jacket removed and 1 inch of insulation stripped from each conductor, The assembly shall be equal to and interchangeable with Joy part No. X8617-24.

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3.8.3.2 Internal wiring. Internal wiring shall be installed in conduit conforming to WW-C-566 or WW-C-581. Conduit outlets shall conform to W-C-586. A ground wire shall be carried as the fourth wire and shall be grounded to the unit in such a manner as to assure an electrical bond between all electrical apparatus and the equipment on which it is mounted. Branch circuit conductors, including wiring internal to starters, breakers, and other electric wiring and devices which carry motor-running current, shall have a current-carrying capacity of not less than 125 percent of the applicable motor-running current. Control subcircuits not carrying motor-running current shall be not smaller than No. 14 AWG wire. Such subcircuits shall include wiring internal to the control panel but not internal to starters, breakers, and other electrical devices. Plug-in connections in branch circuits shall have a positive lock to prevent accidental disconnection. Each wire in the system shall be number coded near each connector.

3.9 Drives.

3.9.1 Chain. The chains used in drive trains shall be in accordance with MIL-STD-421 and MIL-STD-423, or with MIL-STD-424, as applicable. Chain drives shall not have tension adjustment.

3.9.2 V-belt. V-belt drives shall be in accordance with USA Standard B55.1, or Rubber Manufacturers Association, Inc., “Standard Specifications for Drives Using Narrow V-Belts (3V, 5V, and 8V Cross Sections)”. Jack shaft drives may be used. All V-belt drives shall have provision for adjustment.

3.9.3 Gear. Gear drives shall be in accordance with USA Standard B6.6, Class 2 or Class 3. Gear drives shall be totally enclosed and shall run in an oil bath. The enclosures shall be as specified in 3.20.4. Cast iron, when used for gears, shall conform to ASTM A-536, analysis and heat treatment as applicable. The driving pinion shall be steel.

3.9.4 Shafting. All shafting shall be either high carbon steel or alloy steel. The shafting shall be turned, ground, and polished.

3.9.5 Torque arm speed reducers. On torque arm speed reducers, the torque arm shall be attached to the speed reducer housing by means of one or two brackets. Each bracket shall be attached to the speed reducer housing by not less than 2 bolts. If the torque arm is not used to tighten the drive belt or chain, the torque arm shall be at an angle of 90 degrees plus or minus 30 degrees with a line drawn between the output shaft and the point at which the torque arm is attached to the speed reducer. If the torque arm is used to tighten the drive belt or chain, the torque arm shall be at an angle of 90 degrees plus or minus 10 degrees with a line drawn

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between the input and output shafts of the speed reducer. The angular measurement specified herein shall apply when the drive belt or chain is correctly adjusted for tension.

3.10 Revolving scrubber. The revolving scrubber shall consist of a structural steel frame on which is mounted a receiving hopper, a scrubber drum, a water system, and a discharge chute. When the revolving scrubber is assembled for operation, the height of the top of the receiving hopper shall be not more than 16 feet above the ground level.

3.10.1 Receiving hopper. A receiving hopper shall be provided to receive material from a 50-foot portable conveyor and feed the material into the scrubber drum by means of a chute.

3.10.2 Scrubber drum. The scrubber drum shall be not less than 48 inches in diameter by 8 feet in length and shall be constructed of steel plates not less than 3/8 inch in thickness, lined with removable steel liner plates not less than 1/4 inch in thickness. The drum shall be provided with two circumferential steel tires. Each tire shall ride on and be supported by two trunnion rollers. The trunnion rollers shall run on antifriction bearings with waterproof and dustproof housings. The drum shall be set at an axial slope with the horizontal such that the inlet end is higher than the outlet end. The two trunnion rollers supporting the high end of the drum shall be flat-faced. The two trunnion rollers supporting the low end of the drum hall be single-flanged, and the flanges shall prevent uphill movement of the drum by contacting the supporting tire on the drum. Downhill movement of the drum shall be prevented by a thrust roller contacting the supporting tire at the low end of the drum. The thrust roller shall be mounted on a vertical shaft, and shall run on antifriction bearings with waterproof and dustproof housings. The drum shall be equipped with not less than six removable baffles or vanes not less than 3/8 inch thick arranged to propel the materials away from the receiving end and deliver them toward the discharge chute. Removable liners and vanes shall be made of steel conforming to ASTM A242. A supported splash shield shall be provided at the discharge end of the drum to minimize spillage.

3.10.2.1 Drum drive. The scrubber drum shall be driven (rotated) by a 10-horsepower (hp) electric motor through roller chain, gears, or friction drive.

3.10.2.2 Drum bearings. Bearings shall be of the greasable, antifriction type conforming to FF-B-187, FF-B-185, or FF-B-171, type and class as applicable, and shall be of the bearing manufacturer's recommended capacity for the application. Mounting of the bearings in a manner

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other than securing against a shoulder shall be subject to the approval or the contracting officer for each specific application.

3.10.3 Water system. A spray pipe with a diameter of not less than 3 inches and a length of approximately 18 inches shall receive water from a 6-inch header water inlet pipe mounted on the unit and shall have a valve for regulating the flow of water. The connection between the header pipe on the unit and the pump line shall be made by a 6-inch rubber pressure hose equipped with flanges at both ends. The valve shall conform to WW-V-58, class 1, type to suit the application. The water system mounted on the unit shall drain without opening any connections.

3.10.4 Discharge chute. A discharge chute shall be provided for discharging materials from the drum into the hopper of the vibrator screen. The material shall be discharged evenly over the width of the vibrating screen. The discharge cylinder on the scrubber may be considered as the discharge chute if it performs the same function.

3.11 Vibrating screen. The vibrating screen shall be 4 feet wide by 10 feet long, with three decks. The top deck shall have a 1-inch-square-opening, the second deck a 1/2-inch-square-opening, and the bottom deck a 1/4-inch-square-opening screen cloth. The vibrating screen shall consist of a live frame with screen cloths, a vibrating mechanism, suitable vibration isolators, and drive, and may include a stationary frame. Chutes and hoppers for feed and discharge shall be provided.

3.11.1 Stationary frame. When provided, the stationary frame shall be structural steel or castings.

3.11.2 Live frame. The live frame of the screen assembly shall be of steel plates on structural steel members with crossmembers to hold the screen cloths. The frame shall prevent movement of the screen cloth during full-load operation. The crossmembers shall create a longitudinal crown of not less than 1/2 inch in the screen cloth. A molded neoprene cushion shall be provided between the screen cloths and the crossmembers. Screen cloths shall be changeable by using common tools. The frame shall be balanced to provide an even flow of material across the full width of the cloth.

3.11.3 Screen cloth. Screen cloth shall be furnished in 5-foot by full width sections and shall be in accordance with MS39127. Two 5-foot sections of each size shall be furnished in addition to those specified in 3.11.

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3.11.3.1 Wire. All screen cloth shall be made of crimping-grade, oil-tempered standard heavy wire conforming to NSGA Circular No. 80, table 3, grade B, carbon range 0.55 to 0.70, drawn and hardened. Wire size shall be as shown in table I.

3.11.4 Vibrating mechanism. A mechanical vibrating mechanism shall be provided for activating the live frame. When material from the deck or decks of the screen impinges on the activating mechanism, the mechanism shall be equipped with a replaceable, protective cover. The mechanism shall be driven by a 15-hp electronic motor.

3.11.4.1 Shaft bearings. The vibrating screen shall have double-row, self-alining, antifriction shaft bearings. All bearings exposed to dust shall be provided with labyrinth grease and dust seals.

Table I. Wire sizes

Clear opening (minimum)	Wire diameter
(Inch)	(Inch)
2-1/2	1/2
2	7/16
1-1/2	3/8
1	5/16
3/4	1/4
1/2	0.192
3/8	0.162
1/4	0.135

3.11.5 Spray pipes. The vibrating screen shall be equipped with spray pipes above each screen deck. Each spray pipe shall contain a multiple number of spray nozzles directed downward. The spray pipes shall receive water under pressure from the main water header controlled by a valve and shall be provided with valves for regulating the flow or water for each deck.

3.12 Dehydrator. The unit shall be provided with a dehydrator for fines. The dehydrator shall have a single screw 30 inches in diameter and a tub approximately 22 feet in length, set at an angle not more than 20 degrees from the horizontal. The screw shall be driven by a 10-hp electric motor. A means shall be provided to flush water through the drain trough of the tub.

3.12.1 Screw. The dehydrator screw shall be equipped with replaceable, segmental, iron or steel castings or weldments having a hardness of not less than 430 Brinell. The upper end of the screw shaft shall be mounted on thrust-type antifriction bearings enclosed in an oiltight case

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above the water level. The lower end of the shaft shall be mounted in bearings made of fiber, rubber, or other suitable material, and shall be lubricated with fresh water under a pressure of not less than l5 pounds per square inch (psi), or shall be mounted in outboard bearings with a positive means of excluding water and abrasives to prevent wear and failure of components.

3.12.1.1 Drive gears. The screw shall be driven by hardened steel gears in accordance with USA Standard B6.6, class 2. The gears shall be mounted on antifriction bearings and enclosed in an oiltight case.

3.12.2 Feed box and weirs. A steel feed box and adjustable metal weirs shall be provided at the lower end of the tub to receive material from the discharge chute of the bottom deck of the vibrating screen. Baffles shall be provided to minimize turbulence.

3.12.3 Dehydrator bearings. Except for the water-lubricated bearings specified in 3.12.1, all bearings shall be as specified in 3.10.2.2.

3.12.4 Discharge chute. A discharge chute shall be provided at the upper end of the tub. The height of the chute above ground level shall be as specified in 3.7.2.2.

3.13 Piping equipment. The piping equipment to complete the water-supply system shall consist of a suction hose with terminal pipe, elbow, riser, foot valve, and strainer; supporting ponton for connection to the suction flange of and pump conforming to MIL-P-10360; a pressure pipe line with valves and fittings; and a water-waste flume. Provision shall be made to discharge waste water on either side of the unit.

3.13.1 Suction hose. The suction hose shall be 6 inches in diameter by 30 feet long, conforming to ZZ-H-561, grade B, type I, with 6-inch threaded pipe nipples equipped with flanges for bolting of the pump and terminal pipes vulcanized into each end. A terminal pipe, 6 inches in diameter by 10 feet long, equipped with an elbow and a l-foot-long by 6-inch-diameter riser pipe shall be provided.

3.13.2 Foot valve. A foot valve conforming to MIL-V-16121 shall be mounted on the lower end of the riser pipe.

3.13.3 Pressure pipe line. The pressure pipe line shall consist of a 6-inch sprial riveted or welded pipe having a total length of 1,500 feet, made up of pipe sections not to exceed 30 feet in length, with each section provided with quick-detachable, rubber-gasketed joints. Not less than two lengths of pipe shall be provided with 6-inch bolted flanges on one end for connection to the

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6-inch pipe flanges on the unit, the suction hose line, or the pump. The spiral pipe shall be 0.062 inch nominal thickness and shall be coated inside and outside with asphaltum. Valves and fittings in accordance with table II shall be provided to complete the water system to the pressure hose at the plant. Each fitting shall be provided with a quick-detachable rubber-gasketed joint.

Table II. Pipe fittings.

Number required	Description
2	22-1/2-degree elbow
2	45-degree elbow
4	90-degree elbow
1	6-inch pipe, 1-foot long, with nipple and hose bib on one side.

3.13.4 Waste-water flume. The waste-water flume shall consist of 10 sections of semicircular, open-top flume, each 10 feet long and not less than 16 inches wide; and four turn sections of which two shall be 90-degree right and two 90-degree left. All flume sections shall be made of steel 0.05 inch nominal thickness, with bolted lap joints.

3.14 Trailer components.

3.14.1 Frame. The trailer frame shall be constructed to withstand, without damage or permanent deformation, the static and dynamic loads imposed by operation and by being transported as specified in 3.17.

3.14.1.1 Upper fifth wheel and kingpin. The front of the semitrailer shall be constructed so as to be compatible for coupling with and towing by both the military M123, 10-ton truck-tractor and the M52, 5-ton, truck-tractor (see 3.31). The upper fifth wheel shall be of the full-width-type that permits coupling at any angle up to 75 degrees from straight ahead and shall cover a 36-inch-diameter lower fifth wheel. The upper fifth wheel assembly shall conform to ICC 193.70. The plate shall be high tensile steel (not less than 50,000 psi yield point) and shall be not less than 3/8 inch thick. The plate shall include a coupling ramp having not less than 1-1/4 inch vertical height. Unless the coupling ramp is protected by the number 1 crossmember, drain holes shall be provided to prevent any collection of water. A universal kingpin conforming to MS53038, MS53039, and MS53040 with SAE J700 and J848 kingpins, shall be installed not less than 21 inches from the front of the semitrailer. The kingpin assembly shall permit access to allow personnel to reverse the kingpin without removal of other assemblies.

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3.14.1.2 Rear towing components. Towing components shall be furnished on the rear of the semitrailer in accordance with SAE J849 for double trailer application, except as specified herein. The pintle shall conform to MS51118.

3.14.1.3 Front and rear intervehicular connections. The semitrailer shall be furnished with airbrake couplings as specified in 3.14.5.1 and electrical receptacles as specified in 3.14.6 for connecting (1) the truck-tractor to the semitrailer, (2) the dolly to the semitrailer, and (3) other trailers to the semitrailer. The front connections shall be in accordance with SAE J702, and the rear with SAE J849, except as specified herein for military system compatibility. Service brake couplings shall be on the curb side, color coded blue, and emergency brake coupling shall be on the road side, color coded red. Clearance about the couplings and receptacles shall allow operating personnel to connect the truck-tractor, or dolly hoses and harness, to the semitrailer and from the semitrailer to towed trailers without interference with other components. The airbrake couplings shall be protected with spring-loaded covers. The front electical receptacle shall have leads terminating within a splashproof semitrailer electrical nose box or junction boxes (see 3.14.6).

3.14.2 Rear suspension. The rear of the semitrailer shall be mounted on a dual-tire, multiple-axle assembly. Each axle shall be rated by the axle manufacturer at a commercial highway usage capacity rating of not less than 25,000 pounds, but the load on each axle shall not exceed the limit specified in 3.1.2(c). Bearing cones, bearings, and bearing cups (see 3.14.3) shall be furnished by the same bearing manufacturer. The suspension shall be in correct alinement with the frame and with the axles so that the diagonal measurements from the true center of the kingpin to the outer center of each spindle on the axle do not differ in excess of 1/8 inch for every 20 feet or fraction thereof of diagonal distance. Each subsequent axle shall be parallel to the front axle of multiple axle suspensions such that the axle wheelbases on each side do not differ in excess of 1/8 inch. Multiple axle wheelbases shall be not less than 50 inches center-to-center. Suspension mounts shall be installed on the axles in accordance with the suspension and axle manufacturer's recommendations.

3.14.3 Wheel assemblies. The semitrailer shall be equipped with malleable iron or cast steel spoke dual wheels compatible with the axle assembly specified in 3.14.2. Each assembly shall consist of the spoke wheel hub with bearing cups pressed into place, a cast iron brake drum, two rims, a wheel spacer, rim clamps, hub cap, bolts or studs, nuts, and all other accessories necessary to accomodate the tires specified in 3.14.3.3. Wheels and spacers shall be in accordance with SAE J851. The width of the spacer shall provide a minimum of 0.5 inch dual tire bulge clearance under load.

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3.14.3.1 Wheels. The wheels shall be 15-inch, 3- or 6-spoke. Malleable iron wheels shall be of material conforming to SAE 35018, grade A. Cast steel wheels shall be of material conforming to SAE 0030. Rim clamps shall be malleable iron conforming to SAE 35018, grade A. Studs and bolts shall be in accordance with SAE J429, grade 5 or 6, with a minimum tensile strength of 120,000 psi.

3.14.3.2 Rims and brake drums. The rims shall be size 15 by 7.5, 5-degree, three piece type for tube-type tires, and shall accommodate the tires specified in 3.14.3.3. Rims shall conform to the Tire and Rim Association Yearbook. The semitrailer shall be furnished with one spare rim without tire. The location of the assembly shall permit removal and replacement of a spare tire by not more than two men. The spare tire and rim assembly shall not come loose, be damaged, or interfere with any component of the unit during travel or operation. The brake drums shall have a wearing surface of SAE J431 automotive gray cast iron, SAE G3500 or SAE G4000, and shall be of the anti-bell mouth type. The braking surface shall be compatible with the brake lining specified in 3.14.5.

3.14.3.3 Tires and tubes. The tires shall be in accordance with MS35390-47, 11.00 by 15, 16-ply rating, and shall conform to MIL-T-12459. The tires shall be free of major defects when examined in accordance with MIL-STD-1224. The inner tubes shall be in accordance with MS35392-26. Valve caps shall be of metal and shall be equipped with elastomer inner seals.

3.14.3.4 Balancing. Brake drums, hubs, and tire and wheel assemblies shall be inherently in balance to preclude excessive tire wear and eliminate wheel tramp and drumming at the road speeds specified herein.

3.14.4 Chock blocks. Two wood or commercial metal chock blocks with captive chains and two wire cradles shall be furnished. The captive chain shall be wound in a figure-eight manner around the cradle ears and the hook snapped to the lower cradle loop. The length of the captive chain shall permit the chock blocks to be placed ahead of or behind the semitrailer tires. The chock blocks shall be of a minimum size as shown on figure 1 and shall withstand the imposed forces of being run over by the semitrailer. The cradles shall conform to figure 2. Installation of the cradles shall be with the two cradle ears uppermost and the short loops fastened to an approximately vertical, flat surface.

3.14.5 Brake system. The unit shall be equipped with a waterproof, straight air brake system in accordance with ICC Motor Carrier Safety Regulations, sections 193.40 through 193.43 and 193.45 through 193.50, except as otherwise specified herein. The brake system shall include

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service and emergency couplings (front and rear), relay emergency valve, air reservoir, cylinders or chambers, foundation brake assemblies on each wheel, and all other components and plumbing required for a complete, matched airbrake system. All brake system components and plumbing shall be located, installed, and protected to prevent damage or malfunction from ground obstructions and debris when transported, from falling aggregate when operating, and to obtain maximum efficiency and ease of maintenance. Service and emergency couplings shall be furnished and shall conform to MS35746, except that they shall be of the polarized or noninterchangeable type, color coded blue for service and red for emergency. A spring-type, metal dust cover shall be installed on each coupling to prevent contaminants from entering the opening. The dust cover shall not cause interference when another coupling with a dust cover is connected. In addition, metal identification tags conforming to MS53007 shall be installed near each front and rear fixed coupling. The relay emergency valve shall conform to MS53004, except that the rapid bleed-down vent shall not be furnished with these straight airbrakes. The air reservoir shall conform to SAE J10 and shall have an air storage capacity of not less than two times the minimum recommended air volume for truck-trailers as specified in SAE J813. The reservoir shall be equipped with a remote control or a manual, 90 degree twist drain valve, mounted on the low end of the reservoir. The remote control for the drain valve shall be on the curb side of the semitrailer and shall be accessible to personnel from the standing or kneeling position. The foundation brake assemblies shall be not less than 12.25 by 7.5-inch size. The brake lining shall be of the metallic type having nonfading characteristics when subjected to heat and to water immersion. The lining shall be Class Code FE in accordance with Regulation V-3 of the Vehicle Equipment Safety Commission. The lining shall be selected by the supplier and shall be of standard size as listed in the Automotive Data Book of the Friction Materials Standards Institute, Inc. The lining segments shall be not less than 3/4-inch nominal thickness. The brake lining shall be installed on the shoes by the use of hardened steel bolts and nuts, as recommended by the lining manufacturer. All metal surfaces enclosed by the brake drum and the dust shields shall be coated with lacquer-primer conforming to MIL-P-46093 or TT-P-664. Wedges, pins, cams, rollers, and anchors shall be coated with preservative oil conforming to MIL-L-21260, grade 2.

3.14.5.1 Brake lines. The brake lines shall be routed within the protection of the semitrailer framing and shall be secured by cushioned support clamps spaced at intervals not greater than 3 feet. Through-frame connections or grommets used in conjunction with clamps shall be required where lines pass through sheet metal or framing members. The routing and connection locations shall facilitate replacement for the lines from bulk supply. Not less than 5 inches of slack in flexible air hose shall be provided at each of the relay emergency valve connections to

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permit interchange of the various makes of MS53004 valves. The hose ends of these lines shall be identified by markers conforming to MS39020 near each fitting. Brakeline components shall conform to the following:

(a) Flexible air hose - MIL-H-3992, type I, class l or 2.
(b) Rigid air line - SAE J844, copper tubing
(c) Fittings - SAE J512, heavy-duty airbrake, compression type, reusable.

Note: At the supplier's option, flexible air lines conforming to SAE J40, SAE 40R2, type E or F may be furnished in lieu of the flexible air hose and rigid air lines specified in (a) and (b) above.

3.14.6 Vehicular electrical system. The unit shall be equipped with a waterproof, 24-volt vehicular electrical system for service and blackout night travel. Except as specified herein, the electrical system shall be in accordance with the following requirements:

(a) SAE Handbook, Electrical Equipment and Lighting Section.
(b) ICC Motor Carrier Safety Regulations.
(c) TTMA Recommended Practice RP6-2-9.
(d) ATA Specification E-3.
(e) Figure 3.

All components and wiring shall be so located and guarded as to provide protection from vibration, breakage, and damage from falling and flying aggregates. The electrical components shall be in accordance with table III. Item quantities that are not specified shall be furnished as required (AR) for each semitrailer.

Table III. Vehicular Electrical System

Description	Requirements	Quantity per semitrailer
Service lights; clearance side-marker, and identification	MS35423	16 min.
Blackout lights; clearance and side-marker	MS35424	8
Service combination lights	Commercial SAE-SIT, 4-in. dia., waterproof, double contact (SAE J759)	2
Blackout combination lights	MS51330 (Military)	2
Reflectors	MS35387	10
Front trailer receptacle	MS75021-1 with matching spring loaded 4-bolt cover	1

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Table III. Vehicular Electrical System (Con't)

Description	Requirements	Quantity per semitrailer
Semitrailer nose box or double junction boxes	Figure 3, Waterproof type	1 Assy.
Rear trailer receptacle	MS75021-2 with matching 4-bolt spring cover	1
Low tension cable	MIL-C-13486, 14 AWG min.	AR
Insulation	MIL-I-15126 and MIL-I-631	AR
Loom	Polyethylene tubing or SAE J562, oilproof loom	AR
Cushioned support clamps	MS21322, MS21333, and MS21334	AR
Circuit number bands	MS39020	AR
Waterproof friction connectors	MS27142 thru MS27148 and MS33800	AR
Connector and splice lubricant	MIL-S-8660	AR
Connector holder clip	Plastic or metal, spring clip	1 per connector
Threaded plugs and receptacles	Waterproof military type	AR
Terminal connectors	Spade type per SAE J561	AR
Terminal sleeves	Heat shrinkable tubing	AR
Circuit Breakers	MS39062	3
Intervehicular harness (for dolly converter only)	MIL-C-13486/12 with MS75020 plugs on each end mating truck to semitrailer	1

3.14.6.1 Lighting. All lights shall operate as specified herein on 24 volts. All lights shall be provided with shock and vibration protection by location guarding, and through use of resilient mountings to prolong filament life. Front and rear identification, clearance, and side-marker lights that are not protected from vertically falling aggregate shall be cluster-mounted on plates inserted into lamp bracket sockets conforming to SAE J725. These clusters shall be removable for storage in the toolbox during periods or operation. Each cluster shall be equipped with identifying plates, as specified in 3.24.1, to preclude incorrect replacement of the cluster in preparation for travel. The rear service combination lights shall be mounted in industry standard 4-1/2 inch diameter holes. These lights shall be of the flush or recess mounted type having either a sealed unit mounted in an elastomer ring, or a bulb and snap-in lens mounted in an elastomer watertight housing. When furnished, bulbs shall be double contact, indexing base type,

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Trade No. 1662. These lights shall be installed as far apart as possible for turn-signal operation as well as stop and taillight indication. Any legal requirement for the service stoplights to be illuminated whenever the brake system goes into "emergency" shall not be considered applicable. The blackout combination light shall be installed inboard or below the service combination lights. These lights shall be installed and centered behind 6-inch diameter holes and shall be flush with the service combination lights. Shielding shall be provided to prevent damage from road splash.

3.14.6.2 Wiring. The semitrailer wiring shall be routed within the framing members whenever practical. Clamps, grommets, and loom shall be provided to prevent whipping, chafing, or other mechanical damage. Wiring shall be fastened at intervals not to exceed 3 feet. Two or more wires in the same routing shall be bound together into a harness with one-half over-lapping wraps of plastic electrical tape, or covered with tubing insulation. Splices shall be waterproof. Splices shall be either soldered and insulated or mechanical splicing devices shall be furnished at splice points. These splicing devices shall be fastened to the semitrailer members and the interior waterproofed with silicone lubricant. All circuits shall be identified by number coding as specified in figure 3, using circuit number bands near each connection. Friction connectors shall be lubricated with silicone lubricant prior to assembly, and shall be secured to the semitrailer with spring clips to prevent whipping. A waterproof, electrical, semitrailer-type commercial nosebox with insulated terminal strips and terminals; or two junction boxes shall be furnished. The nosebox or junction boxes shall contain connectors for all circuits and the nosebox shall additionally house the three circuit breakers. These components shall be located in the drawbar area. From the connections the harness shall be routed rearward along the frame side member to the unit lights. The service combination lights shall be equipped with friction connectors on the two live, pigtail leads, and a spade connector on the grounding pigtail lead for bolting to the frame.

3.14.6.3 Reflectors. Reflectors shall be furnished and installed as required by ICC regulations. In addition, two amber reflectors shall be installed on the front corners within the same height range as those on the side.

3.14.7 Landing gear and stabilizing jacks. The semitrailer shall be equipped with not less than four screw-action stabilizing jacks and a semitrailer landing gear permanently attached to the chassis. The stabilizing jacks shall conform to GGG-J-51, type I, class 3, diameter of the screw not less than 2-1/2 inches, and rated capacity 15 tons. The jacks shall have steel

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steel ball-bearing caps. The base of the jacks shall be fully enclosed. Detachable pads shall be furnished. Each jack pad shall have sufficient area to reduce ground pressure to less than 100 psi. Provisions shall be made for stowing the pads in the toolbox. The four stabilizing jacks shall fold and be secured for travel. The semitrailer landing gear shall be of the two-speed, telescopic, nonrotating leg type, with mud pads. The landing gear shall be in accordance with the TTMA Recommended Practice RP 4-1-4. Military truck-tractor compatibility for RP 4-1-4 shall be as specified in 3.1.2 (i) for 5-ton M52 and with step extensions installed for the 10-ton M123. Extensions shall be stored on the unit when not in use. Supports for the crank extension shaft and clips to hold the crank in travel position shall be provided. The landing gear shall be protected to preclude the entrance of foreign matter which would impair its functioning or mechanical efficiency. The landing gear shall withstand, without deformation, the combined static and dynamic forces due to its proportion of the total weight imposed and the forces from impact during coupling and uncoupling operations with the emergency breakaway brakes actuated and chock blocks in place.

3.15 Dolly. The dolly shall be furnished by the supplier or will be furnished by the Government as specified (see 6.2).

3.15.1 Government furnished dolly. When the dolly is Government furnished it will be a 4-wheel, 15-ton, M354 dolly conforming to MS500081 (see 6.4).

3.15.2 Supplier furnished dolly.

3.15.2.1 Fifth wheel. A (fixed) lower fifth wheel similar to MS53034 and compatible with the SAE J848 kingpin, shall be furnished. The height of the fifth wheel with semitrailer loaded shall be 58 inches plus or minus 2 inches. Oscillation shall be permitted in a transverse direction only.

3.15.2.2 Lunette and towbar. A lunette in accordance with MS51337 shall be installed in a housing integral with the towbar. The housing shall support the lunette the entire length of the shank. The clearance between the shank and support shall be not greater than 0.035 inch. When the dolly is in towing position, the height of the lunette shall provide compatability with pintles conforming to SAE J849 for Doubles trailer except the pintle height shall be 31 through 37 inches from level ground. The center of the lunette and the centerline of the towbar shall be located within 1/8 inch of the longitudinal centerline of the dolly. The towbar shall be hinged. When the dolly is loaded and the towbar is not locked rigid, the hinge shall permit the lunette to lower without additional load. The towbar shall be of the rigid “A” frame type. When coupled to the semitrailer, the dolly shall turn not less than 110 degrees both to the right and to the left of

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the longitudinal centerline of the semitrailer. The lunette and towbar shall be compatible for towing with military, 5-ton, 6 x 6, M51 dump truck (MS500010) or with a military, 10-ton, 6 x 6, M123 truck-tractor (MS500070) and shall permit the centerline of the truck to be at an angle of not less than 60 degrees to the right or to the left of the centerline of the dolly when in towing position. There shall be no interference between either the truck or the dolly-semitrailer combination.

3.15.2.3 Landing leg. The towbar of the dolly shall be provided with an adjustable gear type landing leg to permit one man to couple and uncouple the lunette and the pintle on the towing vehicle with the dolly loaded. Ground clearance shall be not less than 18 inches when the towbar and landing leg are in travel position.

3.15.2.4 Dolly suspension. The dolly suspension shall include two dual-tired axles as specified in 3.14.2 and 3.14.3 in a single point equalizing tandem axle configuration. The location of the flexible members of the suspension system shall be between the axle and the dolly fifth wheel plate in a manner to provide the oscillation specified herein without hazardous handling characteristics. Tires rims, and wheels shall be interchangeable with those on the semitrailer. A complete air brake system shall be furnished as specified in 3.14.5. The dolly brakes shall be correctly balanced in application pressure and timing to reduce the possibility of jackknifing the unit during stops. The dolly brakes shall have less effectiveness than the rear unit brakes through use of smaller chambers or by installation of a pressure holding or reducing valve in the dolly brake system.

3.15.2.5 Dolly electrical system. The electrical system of the dolly shall conform to 3.14.6 through 3.14.6.3. Service and blackout rear combination lights, only, shall be required and installed.

3.15.2.6 Dolly intervehicular connections. The dolly shall be provided with electrical and airbrake lines to connect the truck systems to the semitrailer systems. All forward intervehicular connections between the dolly and the truck shall extend 36 to 42 inches toward the truck and terminate on the dolly towbar rearward at least 18 inches, measured from the center of the lunette eye. All rearward intervehicular connections between the dolly and the semitrailer shall be supported on a truck-tractor type spring pole. The intervehicular connection shall have a minimum length of 10 feet, measured from the dolly frame to the connectors and shall allow the dolly to turn as specified in 3.15.2.2. Plug clips and rigid mount dummy couplings, shall be mounted on the dolly and spring pole to hold both the forward and the rearward intervehicular connections during temporary storage and travel, when not in use. Electrical and airbrake components shall be as specified in 3.14.1.3 and shall include hose spring guards and dust covers

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on the airbrake coupling hose ends. Forward, intervehicular connections shall be routed so that dropping of the tail gates of the military dump and cargo trucks shall not cause damage.

3.15.2.7 Safety chains. Two safety chains fitted with grab hooks shall be furnished. The safety chains shall extend 36 to 42 inches beyond the center of the drawbar lunette eye. The safety chains shall be in accordance with ICC Motor Carrier Safety Regulations, except that the safety-chain hooks shall fit the shackles and attachments of the towing vehicles and of the semitrailer (see 3.14.1.2).

3.16 Auxiliary components. Access ladders, platforms, and handholds shall be provided to permit servicing and inspection of all components.

3.17 Transportability. The unit shall withstand impact forces encountered in vehicular, rail, and air transport as specified herein without damage or permanent deformation.

3.17.1 Vehicle mobility. The unit shall meet the performance characteristics when coupled to its dolly converter and towed by a prime mover, such as the 5-ton, 6 x 6, M51 dump truck shown on MS500010 or crawler tractor(s), and when directly coupled as a semitrailer to a 5-ton, 6 x 6, M52 truck-tractor shown on MS500008 and to a 10-ton, 6 x 6, M123 truck-tractor shown on MS500070. The vehicle mobility requirements specified in 3.17.1 through 3.17.1.5 shall apply when the height of the unit has been reduced for highway transport as specified in 3.1.2.

3.17.1.1 Highway operation. The unit shall travel over smooth, hard-surfaced primary highways at speeds up to 30 mph.

3.17.1.2 Cross-country operation. The unit shall negotiate unimproved roads and open, rolling, and hilly cross-country terrain at speeds up to 10 mph. The unit shall traverse grades up to 40 percent; mud, sand, and snow; and transport incident to intended operation sites, when towed by crawler tractor(s). Without prior preparation, the unit shall ford hard-bottomed bodies of water up to 30 inches in depth without damage or adverse effects because of immersion.

3.17.1.3 Trailing and turning ability. The unit shall trail the towing vehicle without weaving or sidesway. The unit shall follow the truck-tractor without exceeding 75 percent of the ICC tracking deviation, as specified in 193.70 of the ICC Regulations, when towed over a paved surface. Jackknife turns up to an angle of 90 degrees for truck-tractor semitrailer combination

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and turns up to 60 degrees for truck-dolly semitrailer combination as measured from straight-ahead to the right and to the left shall not cause damage to or interference between the prime mover, dolly, or unit.

3.17.1.4 Braking ability. The combined service brakes of the prime mover and unit shall control, decelerate, and stop the combination in accordance with ICC Regulations, except that the brake application and stopping distance shall be accurately measured from initial movement of the brake pedal to the stopping point. The emergency breakaway system shall provide a controlled, emergency brake application that shall stop and hold the unit without hazardous unbalance when actuated by disconnect, breakage, or leakage of the emergency line. With the brake system of the unit stabilized at a maximum working pressure of 100 psi, air leakage within the dolly-semitrailer brake system tested in accordance with TTMA3-1-12 shall not exceed 0.5 psi per minute with the brakes released, nor exceed 1.0 psi per minute with the brakes applied.

3.17.1.5 Suspension system. The suspension system of dolly and unit shall withstand the imposed loads without evidence of overload, wear, malfunction, or permanent deformation when subjected to the loadings and the operating conditions specified herein. When the dolly is coupled to the semitrailer, the articulation shall provide not less than 6 inches difference in elevation between opposite wheels on each axle without loss of tire-to-ground contact. In addition, not less than 6 inches articulation shall be required between all combinations of diagonally opposite wheels on multiple axle suspensions without loss of tire-to-ground contact. The 6 inches of articulation shall be 3 inches up and 3 inches down.

3.17.2 Railroad. Sectionalization for rail transportation shall be within the dimensional limits shown on figure 4, except that item length shall be not more than 38 feet 10 inches. Sectionalization and reassembly shall each be accomplished in not more than 6 hours without cutting or welding.

3.17.3 Air Transport. Sectionalization for air transport shall be within the dimensional limits shown on figure 5. Sectionalization and reassembly shall be accomplished as specified in 3.17.2.

3.17.4 Lifting and tiedown attachments. All sections separated to meet the requirements of 3.17.2 and 3.17.3 shall be provided with lifting and tiedown attachments conforming to 3.22 and 3.23, respectively.

3.18 Feeder assembly. The feeder assembly shall consist of a surge hopper with removable grizzly, and an electric-motor-driven reciprocating plate feeder with adjustable gate.

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3.18.1 Surge hopper. The surge bin hopper shall be mounted on the plate feeder, shall have a capacity of not less than 2 cubic yards, and shall be equipped with a demountable sloping bar grizzly.

3.18.1.1 Hopper grizzly. The demountable grizzly shall be mounted on the top of the surge hopper. The grizzly bars shall be high-carbon, heavy-duty bars spaced to provide 3- by 12-inch openings.

3.18.2 Reciprocating plate feeder. The reciprocating plate feeder shall have a maximum capacity of not less than 150 tph of dry material and shall discharge material without spillage onto a belt 24 inches wide. The feeder shall be driven by 5-hp electric motor.

3.18.2.1 Feeder mechanism. The reciprocating plate shall be supported on rollers mounted on antifriciton bearings. The reciprocating plate, rollers, shafts, bearings, and hoppers shall be supported by structural members and connections. All bearings shall be antifriction with labyrinth grease and dust seals; however, the eccentric bearings may be of the babbitt type. The plate feeder may have an adjustable stroke.

3.18.2.2 Feeder bearings. All antifriction bearings shall be in accordance with FF-B-171, FF-B-185, or FF-B-187, type and class as applicable. Mounting of the bearings in a manner other than securing against a shoulder shall be subject to the approval of the contracting officer for each specific application.

3.18.2.3 Adjustable gate. The feeder shall be provided with a gate having a height-adjusting mechanism covering the range of capacity from 50 tph to 150 tph. Changing the feeder stroke to assist feed-rate variation is permissible, but not less than 50 percent of the feed-rate variation shall be provided by the gate mechanism. The gate shall rise automatically to permit passage of oversize rock and shall then return to the adjusted setting.

3.18.3 Mounting. The feeder shall be provided with four legs which shall be detachable from the feeder for transport. When assembled for discharge onto a portable conveyor, the height of the discharge chute shall be not less than 48 inches from the ground.

3.19 Electromagnetic interference. The electromagnetic interference emission and susceptability characteristics of the unit shall conform to MIL-STD-461, class IIB.

3.20 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

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3.20.1 Lubricants. The unit shall operate as specified herein when lubricated with military lubricants (see 6.5).

3.20.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal of adjustment of accessories or parts. Where oil holes are applicable, they shall have spring-loaded covers. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility. Grease fittings on electric motors shall be removed, after being lubricated, and replaced with threaded, metal plugs.

3.20.3 Pressure release device. An automatic pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.20.4 Enclosure with integral reservoir of lubricant. Enclosures such as gear cases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, finger holes, or sight holes to determine the level of the lubricant. Each enclosure shall be equipped with a magnetic drain plug not less than 1/2-inch pipe size conforming to MS35844 or MS49006, and with a means for filling the enclosure with lubricant. The drain plug shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the unit is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to the ground. Accessibility to the drain plug, the filling means, and the lubricant-level checking device shall be obtained without the removal or adjustment of accessories or parts, except that panels and plates equipped with hand-operable, quick-disconnect, fasteners may be removed.

3.20.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases conforming to MIL-G-10924 or MIL-G-23827, as applicable. The unit shall be run-in (broken-in) and delivered with these greases. The bearings shall be cleaned before lubrication because military greases are not normally compatible with other greases. A tag attached in a conspicuous location shall indicate which military grease has been used.

3.20.6 Oil lubrication. Unless otherwise preserved for packaging (see 5.1), all components shall be filled to the operating level with grades of military oils designated for use in the

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temperature range specified (see 6.2). A tag attached in a conspicuous location shall indicate the temperature range and grade or lubrication used.

3.21 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with varnish conforming to MIL-V-173 except that 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish shall be used in lieu of salicylanilide, the specified fungitoxicant:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component and element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which could interfere with the adherence or functioning of the varnish.

3.22 Lifting attachments. Each unit and each sectionalized component shall be provided with permanently attached lifting attachments (see 3.17.2 and 3.17.3). The attachments shall enable the unit and each sectionalized component to be lifted in its normal traveling or operating position. The attachments shall be located so that not less than 1-inch clearance is maintained between the slings and all exterior parts of the unit or each sectionalized component being lifted. Spreader bars are permitted. Each attachment shall be of sufficient strength to carry not less than 2-1/2 times the normal load imposed on the attachment. Attachments for multiple slings shall be located so that the attached slings will converge not more than 19 feet above the lowest extremity of the load. Attachments shall be positioned so that the lifting strain will be in line with the longitudinal axis of the eye of the attachment. The inside diameter of the eye shall be not less than 3 inches. Attachments shall be attached to members of sufficient strength to withstand stresses in the amount and direction of pull specified for the attachment. Where required, reinforcements shall be used to meet the requirements specified herein. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.23.

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3.23 Tiedown attachments. Each unit and each sectionalized component shall be provided with the integral attachments to permit tiedown of the unit or section to the floor or deck of the transportation medium (see 3.17.2 and 3.17.3). The tiedown attachments shall withstand dynamic thrusts encountered during transport, and static loads of 8G-forward, 8G-rearward, 2-1/2G-upward, and 1-1/2G-sideward where G equals the maximum shipping weight of the unit or each sectionalized component. Attachments shall be rounded and beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments when such attachments meet requirements specified in 3.22.

3.24 Identification marking. The unit shall be identified in accordance with MIL-STD-130. The marking shall be applied to the unit on plates conforming to MIL-P-514, type I, style 1, composition C, of type 1, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.24.1 Shipping data plate. Shipping data plate shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and in addition, shall show the silhouette of the unit in transport position indicating the center of gravity and the location and capacity of the lifting and the tiedown attachments. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.24.2 Instruction plates. Instruction plates shall be furnished and shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material, dimensions to suit the application. These plates shall indicate sequence for start up, shut down, and normal operation; production capacity; and all inherent dangers which can stem from unsafe operation or an unsafe operating position. The safety, warning, caution, and notice plates shall be in accordance with USA Standard Z 35.1, except that the size may be reduced to be compatible with the available space. All plates shall be securely attached to the unit with machine screws, bolts, and nuts, or rivets, and shall be furnished and mounted by the supplier.

3.24.2.1 Warning plates. The following warning plates shall be furnished:

(a) A plate placed on the outside of the distribution panel (see 3.8.2.4), near main power inlet with the following wording:
"Warning. Install ground rod and lead by bolting ground wire to grounding stud on the unit frame before this unit is operated. A ground rod and lead should also be installed at power source (generator)."

(b) A plate placed adjacent to grounding stud or bolt (3.8.3.1) with the following wording:
"Attach ground wire here."

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(c) Two plates, one on each side of the semitrailer at the kingpin, each with the following wording:

"Warning. This kingpin is reversible, having 2-inch and 3-1/2-inch throat diameters. When coupling this semitrailer to the dolly or to a truck-tractor, be sure to use the correct end of the kingpin."

(d) One plate located in a conspicuous place on the dolly having the following wording:

"Warning. Do not connect this dolly to any semitrailer unless a 3-1/2-inch throat diameter kingpin is in place on the semitrailer."

3.24.2.2 Wiring diagram. A plate containing a schematic wiring diagram shall be mounted near the operator's position. The diagram shall show the physical location all wiring interconnections in the same relationship as when they are installed. Identification marking of wires, instruments, control devices, and connections shall be shown and shall coincide with markings on the items installed. Electrical symbols shall be in accordance with MIL-STD-15, WWhen additional electrical symbols are used for items not in MIL-STD-15, the meaning of the symbol shall be defined by a note on the diagram.

3.25 Treatment and painting. The portions of the unit normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.26 Vehicle marking. The marking or the semitrailer shall be in accordance with MIL-STD-642. The total weight and reduced weight for highway transport or the unit (see 3.1.2), and the weight of each sectionalized component shall be marked on the side in such a manner as to be discernible. The prescribed tire pressure shall be marked near the tires. Lifting and tiedown attachments shall be marked "LIFT HERE" or "TIEDOWN HERE", as applicable. Marking of lifting and tiedown attachments for sections shall include the word: "SECTIONALIZATION".

3.27 Technical publications to accompany the unit. Such technical publications as are specified shall be furnished (see 6.2).

3.28 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished (see 6.2).

3.29 Toolbox. The unit shall be provided with one or more weatherproof toolbox(s) made of steel not less than 0.071 inch thick. The toolbox lid shall be mounted on hinges conforming to MS35829 with a 0.125-inch-diameter brass pin. The toolbox lid shall open to at least 90 degree

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from the closed position. The lid shall remain in the fully open position either by its own weight or by detent holding devices. Two tension-type fasteners with strikers shall be provided to retain the lid in the closed position. The fasteners and strikers shall be attached to the toolbox by either bolts or rivets. A hasp which will acommodate a padlock having a 3/8 inch shackle shall be provided for locking the toolbox. The toolbox shall be fastened to the unit in a protected position and the top or the box shall be not more than 4 feet from the ground. A drain hole with rubber plug shall be provided in the bottom of the toolbox. The toolbox shall be of a size to hold the emblem and blade (see 3.6), the ground rods (see 3.8.3.1), the jack pads (3.14.7), the tools specified (see 3.28), and the running light clusters (see 3.14.6.1). The toolbox shall be marked to indicate the items it contains.

3.30 Items to be furnished with the unit. No shipment of units shall be made unless repair parts, maintenance tools, and technical publications are included with the shipment or unless approval for shipment of the units without such items has been received from the contracting officer.

3.31 Government property.

3.31.1 Government-furnished property. When specified (see 6.2), one dolly, 4-wheel, 15-ton, M354 will be furnished for each unit.

3.31.2 Government-loaned property. The following property in the quantities indicated will be loaned by the Government (see 6.6):

Item No.	Description	Identification	Qty. f/ea Unit
1	M-52 - Truck Tractor, 5 Ton, 6 x 6	MIL-T-740	1
2	M123 - Truck Tractor, 10 Ton, 6 x 6	MIL-T-62012	1
3	M51 - Truck, Dump, 5 Ton, 6 x 6	MIL-T-740	1

3.33 Workmanship. All parts components, and assemblies of the unit including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux and other harmful extraneous material. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges and corners are required.

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3.32.1 Steel fabrication. Steel used in the fabrication of the unit shall provide original quality surface finish and shall be free from kinks and sharp bends. Steel having eroded surfaces is not acceptable. The forming of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true, and all sharp edges and burrs shall be removed. Burned surfaces of flame-cut material shall be free of burrs, sharp edges, and slag. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly except where heat treatment is required. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.32.2 Welders and welding.

3.32.2.1 Welder qualification. Welder qualification shall be in accordance with MIL-STD-1261.

3.32.2.2 Welding. Welding of steel shall be in accordance with MIL-STD-1261, and as specified herein except that joint welding procedures are not required for class 1 and class 2 steels.

3.32.2.3 Aluminum welding. Aluminum welding shall be done by the metal-inert-gas method or any other method that will provide equivalent mechanical properties of the filler metal to base metal combinations. All aluminum welds shall be made by the automatic-machine method. All welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. Aluminum welds may be ground, filed, wire brushed, or chipped but shall not be hammered. Preheating for welding is permissible provided the temperature does not exceed 400° F. for a total time of 30 minutes. No process requiring the use of welding flux shall be used.

3.32.3 Brazing. Brazing of steels, copper alloys, and nickel alloys shall conform to MIL-B-7883. Brazing of aluminum and aluminum alloys shall not be permitted.

3.32.4 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or any other defect which might render the casting unsound for its intended use.

3.32.5 Bolted connections. Bolt holes shall be accurately formed or drilled and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. All bolts, nuts, and screws shall be correctly torqued. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt.

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3.32.6 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.32.7 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein.

3.32.8 Forgings. Forged components shall be in accordance with the Forging Industry Handbook of the Forging Industry Association.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The unit shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The unit shall be tested as specified in 4.6.2 through 4.6.2.6. Failure of any test shall be cause for rejection.

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4.4 Initial production inspection. When an initial production unit(s) is specified (see 3.3), units will be selected at random by the Government from the first five units being produced from production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2 through 4.6.2.6 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production unit shall not exclude the remaining units from the quality conformance inspected and acceptance provisions specified in section 4.

4.4.1 Inspection failure. Failure of an initial production unit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production units and shall be cause for refusal by the Government to continue acceptance of units until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on units previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all units accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each unit shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each unit shall be tested as specified in 4.6.2.7. Failure of the test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The unit shall be examined for the following defects (items 114 through 116 shall be examined while the unit is operating):

101. Material not as specified.
102. Parts or components missing or do not function.
103. Assembly is incomplete or incorrect.
104. Safety devices or guards missing.
105. Workmanship not as specified.
106. Ease of maintenance not as specified.
107. Dimensions or weight not as specified.

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108. Treatment and painting not as specified.
109. Identification or other marking, operating, and safety instructions or manuals missing.
110. Lubrication devices not accessible.
111. Brake preservation not as specified.
112. Brake system not as specified or does not function.
113. Electrical system not as specified, does not function, or is damaged.
114. Oil or grease seals leaking.
115. Bearings or gear boxes overheating.
116. Shaft misalinement.
117. Center-of-gravity height not as specified in 3.1.2 (m).
118. Electromagnetic interference suppression components not identical with approved model.

4.6.2 Tests. Prior to test, the unit shall be lubricated with military oils and greases specified in 3.20. Oils shall be those designated for use in the ambient temperature at which the test is being performed.

4.6.2.1 Washing capacity.

4.6.2.1.1 Test conditions.

(a) Operating time. The production capacity of the unit shall be computed on the basis of the net operating time which is defined as gross operating time less allowable down time for the following:

(1) Delay due to lack of material delivery to the unit.
(2) Normal machinery maintenance. (Down time due to any design deficiency shall not be considered normal maintenance.)
(3) Conditions considered by the contracting officer to be beyond the control of the supplier.

(b) Aggregate material. The aggregate delivered to the unit shall be material as specified in 3.1.1.
(c) Tolerance. The sized material produced shall be within the following tolerances:

(1) For the l-inch, 1/2-inch, and 1/4-inch screens, not more than 10 percent of the material in the over-product shall pass a screen with the same size opening and not more than 10 percent of the material in the through-product shall be retained on a screen with the same size opening.
(2) The 1-1/2-inch to 1/4-inch material shall have not more than 1 percent clay lumps when tested in accordance with SS-R-406, method 205.0.

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(3) The 1/4-inch-minus product shall have not more than 1 percent of material finer than No. 200 mesh sieve when tested in accordance with SS-R-406, method 202.11.

4.6.2.1.2 Capacity. Operate the unit for 40 hours net operating time, washing and screening the material specified in 4.6.2.1.1 (b) using the feeder assembly and a portable conveyor. Cumulative daily runs will be permissible in performing the test. Four sizes of material shall be produced: (a) 1-1/2-inch to 1-inch; (b) 1-inch to 1/2-inch; (c) 1/2-inch to 1/4-inch; and (d) 1/4-inch-minus. Production of less than 75 tph as specified in 3.1.1 or evidence that any part or component does not function as specified herein shall constitute failure of this test. Nonconformance to 3.18.2 and 3.18.2.3 shall constitute failure of this test.

4.6.2.2 Transportability tests. Tow the unit, with components removed to meet the requirement of 3.1.2 (f), as follows:

(a) 250 miles over highways with the M52 truck-tractor at speeds up to 30 mph.
(b) 10 miles cross country with the M123 truck-tractor at speeds up to 10 mph.
(c) 250 miles over highways and 10 miles cross country with the dolly specified in 3.15 and the M51 dump truck ballasted as required to provide traction.
(d) Before any highway and cross-country tests are conducted, measure all critical mounting points on the unit. After all highway and cross-country tests are completed, check the measurement of critical mounting points and inspect all welds.
(e) Tow over 6-inch high obstacles such that all left and right combinations of wheels have simultaneously passed over the obstacles. Observe tire-to-surface contact, evidence of binding, or interference. After returning the unit to level ground, measure the diagonals and parallels for evidence of axle misalinement.
(f) Tow over V-ditches not less than 36 inches wide and 18 inches deep. Negotiate the ditches both perpendicular and at an angle of 45 degrees to the direction of travel.
(g) Measure and record the deviation or off-tracking of the semitrailer with respect to the dolly and to the prime movers when the semitrailer is towed straight ahead on a paved surface.
(h) Ford the semitrailer through a body of water not less than 30 inches deep. While in the water turn off the lights.
(i) Demonstrate brake system conformance to ICC regulations.
(j) With the truck-tractor only, tow the semitrailer up to 30 mph and apply the brakes to produce a deceleration rate of not less than 10 feet per second per second, until

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slowed to 10 mph. Immediately accelerate to 30 mph. Repeat this snubbing cycle for a total of 5 cycles and make a panic stop on the 6th cycle.

(k) At 10 mph, cause the unit to go into "emergency", simulating a breakaway from the dump truck and dolly-semitrailer.

(l) Test the brake system in accordance with TTMA recommended practice RP3-1-12.

During the above tests, lights shall be on and turn signals actuated on turns. Upon completion of the travel, remove the rims and wheels and examine the entire unit, including the rims and wheels. Damage to or permanent deformation of any part of the unit, change in critical mounting point dimensions, evidence of any weld being defective, inability to complete any portion of the test, incompatibility with any towing vehicle, or operating difficulty shall constitute failure of this test. Nonconformance to 3.14, 3.15, and 3.17 shall constitute failure of this test.

4.6.2.3 Railroad and air transport. Sectionalize and reassemble the unit. Nonconformance to 3.17.2, 3.17.3, or 3.17.4 shall constitute failure of these tests. The tests for sectionalization and for lifting and tiedown attachments may be accomplished in conjuction with the tests specified in 4.6.2.4 and 4.6.2.5.

4.6.2.4 Lifting and tiedown attachments. Apply the static load specified in 3.22 for lifting attachments and 3.23 for tiedown attachments to the attachments after the attachments are mounted on the unit or section, as applicable. The direction of the applied load shall be in the direction in which the load is normally applied. Weld failure or permanent deformation of the attachment or of any component of the unit or section shall constitute failure of this test.

4.6.2.5 Tiedown attachments (dynamic). Anchor each section of the unit independently, as sectionalized for rail transport, to a flat car using only the tiedown attachments. Use no blocking. Hump the flatcar at a speed of 7 miles per hour against five box cars with their brakes set. Repeat the test with the opposite end of the section in the direction of the five cars. Any permanent deformation or breakage of the tiedown attachments or parts of the section shall constitute failure of this test.

4.6.2.6 Electromagnetic interference. The preproduction model shall be tested to determine conformance to 3.19. The supplier shall furnish to the contracting officer the test report required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.6.2.7 Production. Operate each unit for a period of 1 hour without processing material. Test each brake system in accordance with TTMA RP3-1-12. Any operational difficulties,

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breakage, or permanent deformation shall constitute failure of this test. Operate brakes; nonconformance to 3.17.1.4 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select units at any time during the contract production period and subject these units to the examination specified in 4.6.1 and the tests specified in 4.6.2 through 4.6.2.6 to determine that the quality of the selected unit is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Units will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production units.

4.7.1 Inspection failure. Failure of any unit when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of units until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of preparation for delivery.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. Examine the preproduction pack for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.1.2 Tests. The preproduction pack of crated components for level A shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, appendix A. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of not less than 10 miles or more than 11 miles per hour. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the components shall constitute failure of the preproduction pack and shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level S-2.

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4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

119.	Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall be considered one defect.
120.	Sectionalization and disassembly not as specified for level A or B.
121.	Parts not matchmarked to facilitate reassembly for level A or B.
122.	Cloth tags for matchmarking and lubrication information not waterproofed for level A.
123.	Matchmarking and lubrication information tags not attached for level A or B.
124.	Unpainted exterior metal surfaces not coated with preservative for level A or B.
125.	Semitrailer chassis and dolly components not preserved in accordance with the referenced document for level A or B.
126.	Gears not preserved as specified for level A or B.
127.	Drive chains not preserved as specified for level A or B.
128.	Drive belts not removed or released from tension for level A or B.
129.	Exposed threaded surfaces of loose piping equipment not protected as specified for level A or B.
130.	Openings into electrical components not sealed as specified for level A.
131.	Cables not secured on reels for level A or B.
132.	Maintenance tools not preserved and packaged in accordance with referenced document for level A or B.
133.	Intermediate packaging not as specified for level A or B.
134.	Packaged and unpackaged components secured to the unit interfere with lifting or towing the plant for level A or C.
135.	Strapping not zinc coated for level A.
136.	Blocking, bracing, and anchoring not in accordance with the referenced documents for level A.
137.	Marking illegible, incomplete, or incorrect.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2) to prove, prior to starting production packaging and packing, that the applied preservation, packaging, packing and marking comply with the preparation for delivery requirements of this specification (see 6.7). Examination and tests shall be those specified herein.

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5.2 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Sectionalization and disassembly. The unit shall be sectionalized to meet the applicable requirements of 3.17.2 or 3.17.3. In addition, components of the unit and sections vulnerable to damage, pilferage, and loss, and protruding parts that increase cubage and that can be removed and reassembled without special tools shall be removed. Removed bolts, nuts, pins, and washers shall be replaced in one of the mating parts and secured to prevent their loss.

5.2.1.2 Matchmarking. To facilitate reassembly, removed components and sections shall be matchmarked. Large sections and components shall be matchmarked identically by stencilled letters or numerals with lusterless white enamel, overcoated with varnish conforming to TT-V-121; or identically matchmarked with paint conforming to MIL-P-13983. Small components removed shall have matchmarking information on cloth tags conforming to UU-T-81, type A, or on metal tags. The cloth tags shall be waterproofed in accordance with MIL-STD-129. The tags shall be securely attached to the components so as to prevent damage to the preservation or to the components.

5.2.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2.1.4 Unpainted surfaces. Components of the unit such as chutes, flumes, sprockets, shafting, spray bars, springs, chock blocks, safety chains, coupling pins, towing hooks, bolts, nuts, pins, and washers, any other surface exposed by sectionalization or disassembly, and any other similar unpainted exterior metal surface not definitely specified herein requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative.

5.2.1.5 Trailer chassis and dolly components. The components of the semitrailer chassis and dolly shall be preserved and packaged in accordance with the level A (mobile) requirements of MIL-STD-281.

5.2.1.6 Gears.

5.2.1.6.1 Exposed gears. Exposed gears shall be coated with type P-l preservative or with primer conforming to TT-P-664.

5.2.1.6.2 Enclosed gears. Gears operating on lubricating oil (SAE 10, 30, or 50) shall have the housing filled to the operating level with type P-10, grade 1, 2, or 3 preservative as applicable

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and tagged to indicate: "The preservative is good for operation. Do not drain until the first required lubrication change." Gears operating on gear lubricant (SAE 80 or 90) shall be filled to the operating level with lubricant conforming to MIL-L-2105, grade as applicable, and tagged to indicate: "The lubricant is good for operation." Gears not operating on lubricating oil (SAE 10, 30, or 50) or lubricant (SAE 80 or 90) shall be filled to the operating level with the approved lubricant required for operation. The gear housings shall be tagged to indicate: "The housing is filled to the operating level with the applicable-type lubricant." Tags shall conform to UU-T-81, type A. Tags and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.2.1.7 Drive Chains.

5.2.1.7.1 Exposed chains. Exposed drive chains shall be coated with enough type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. The excess preservative shall be drained, and the entire chain shall be coated with type P-l preservative. Removed chains shall be coiled, and the coil shall be secured with soft, annealed wire and wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured with tape conforming to PPP-T-60, class 1. Sprockets and other unprotected surfaces of the drives shall be coated with type P-1 preservative.

5.2.1.7.2 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified in 5.2.1.6.2 for enclosed gears and gear housings.

5.2.1.8 Shafts and universal joints. Exposed surfaces of shafts and universal joints shall be coated with type P-1 preservative.

5.2.1.9 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1. Unpainted pulley grooves shall be coated with primer conforming to TT-P-664.

5.2.1.10 Piping equipment. Loose piping equipment required to complete the water-supply system shall be preserved in accordance with MIL-P-116, method I or III as applicable. Exposed threaded surfaces shall be protected against damage by the use of metal or plastic thread protectors.

5.2.1.11 Electrical components.

5.2.1.11.1 Openings. Openings into the motors, starters, switches, controls, junction boxes, power receptacles, lamps, and ends of the intervehicular cable, and main feeder cables shall be sealed with tape conforming to PPP-T-60, class 1.

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5.2.1.11.2 Cable reels. Unpainted exterior metal surfaces of the cable reels shall be coated with type P-1 preservative. The main feeder cables shall be wound on their reels and secured.

5.2.1.11.3 Ground-rod assemblies, clamps and wires. The ground-rod assemblies, clamps, and wires shall be placed in the container provided.

5.2.1.12 Technical publications. Techanical publications shall be preserved in accordance with MIL-P-116, method IC-1, and placed in the toolbox.

5.2.1.13 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.1.14 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with the level A requirements of MIL-H-15424 and placed in the toolbox.

5.2.1.15 Intermediate packaging. Packaged and unpackaged components and repair parts that cannot be secured to the unit or sections shall be intermediate packaged in close-fitting boxes conforming to PPP-B-621, class 2, style optional; or PPP-B-601, overseas type, style optional.

5.2.2 Level B. The complete units shall be preserved and packaged as specified for level A, except as follows.

5.2.2.1 Tags. Cloth tags specified in 5.2.1.2 and 5.2.1.6.2 shall not require waterproofing.

5.2.2.2 Trailer components. The components of the semitrailer chassis and dolly shall be preserved and packaged in accordance with the level B (mobile) requirements of MIL-STD-281.

5.2.2.3 Exposed chains. Exposed drive chains shall be preserved as specified for level A, except only type P-3 or P-9 preservative shall be applied.

5.2.2.4 Drive belts. Drive belts shall be released from tension.

5.2.2.5 Electrical component openings. Openings into electrical components specified in 5.2.1.11.1 shall not require sealing with tape.

5.2.2.6 Intermediate packaging. Intermediate packaging shall be as specified for level A, except boxes shall be domestic type and class as applicable.

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5.2.3 Level C.

5.2.3.1 Repair parts, maintenance tools, and technical publications. Repair parts and technical publications shall be preserved and packaged as specified for level A. Maintenance tools shall not require preservation but shall be placed in the toolbox as specified for level A.

5.2.3.2 Other components. Preservation and packaging of all other components of the unit shall afford protection against deterioration and damage from the supplier to the initial destination. The supplier's standard commercial practice is acceptable when it fulfills these requirements.

5.3 Packing. Packing shall be level A or c as specified (see 6.2).

5.3.1 Level A. The units shall be prepared for mobile shipment. When sufficient space is available, packaged and unpackaged components and sections shall be placed and secured on the unit in any available space that will not interfere with the lifting or mobility of the unit. The components shall be secured to the unit with zinc-coated strapping conforming to QQ-S-781 or QQ-S-790. Packaged and unpackaged components and sections that cannot be secured to the unit shall be packed in crates conforming to PPP-C-650, style A, or MIL-C-3774, nailed assembly, skid-type base. Blocking, bracing, and anchoring shall be in accordance with the applicable crate specification and with MIL-STD-1186. Crate components shall be shipped at the same time as the mobile unit.

5.3.2 Level C. The complete unit shall be prepared for shipment in a manner to assure carrier acceptance and safe delivery to destination at lowest rating. The unit shall be prepared for mobile shipment. When sufficient space is available, packaged and unpackaged components and sections shall be placed and secured on the unit in any available space that will not interfere with lifting or mobility. Packaged and unpackaged sections that cannot be secured to the unit shall be packed in crates complying with Uniform Freight Classification Rules or National Motor Freight Classification Rules. Crated components shall be shipped at the same time as the unit.

5.4 Marking. Packages, shipping containers, and mobile plants shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The washing and screening unit is intended for use in screening and washing rock aggregate as a part of a 75 ton per hour rock crushing and screening plant.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) When an initial production unit is required (see 3.3).
(d) Number of units required for initial production inspection (see 3.3).
(e) Whether the dolly shall be furnished by the supplier or will be furnished by the Government (see 3.15 and 6.4).
(f) Grade of military oil required to service washing and screening unit (see 3.20.6).
(g) Technical publications required (see 3.27).
(h) Applicable list of repair parts and maintenance tools (see 3.28).
(i) Time frame required for submission of preproduction pack (see 5.1).
(j) Levels of preservation, packaging, and packing (see 5.2 and 5.3). Level B preservation and packaging is intended to provide economical but limited protection and should be specified only when it is determined the washing and screening units will be held in covered storage.

6.3 Preproduction model. Any changes or deviations of production units from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish units conforming to this specification.

6.4 Government-furnished property. The contracting officer should arrange to furnish the M354 dolly when required (see 3.15.1 and 6.2).

6.5 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the washing and screening unit covered by this specification as contained in the Federal Supply Catalog, DOD Section, Identification List C9100-IL for FSC Group 91.

6.6 Government loaned property. The contracting officer should arrange to loan the property listed in 3.31.

6.7 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the units in accordance with this specification.

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Air Force – 84

Review activities:

Army – EL
Air Force – 85

User activities:

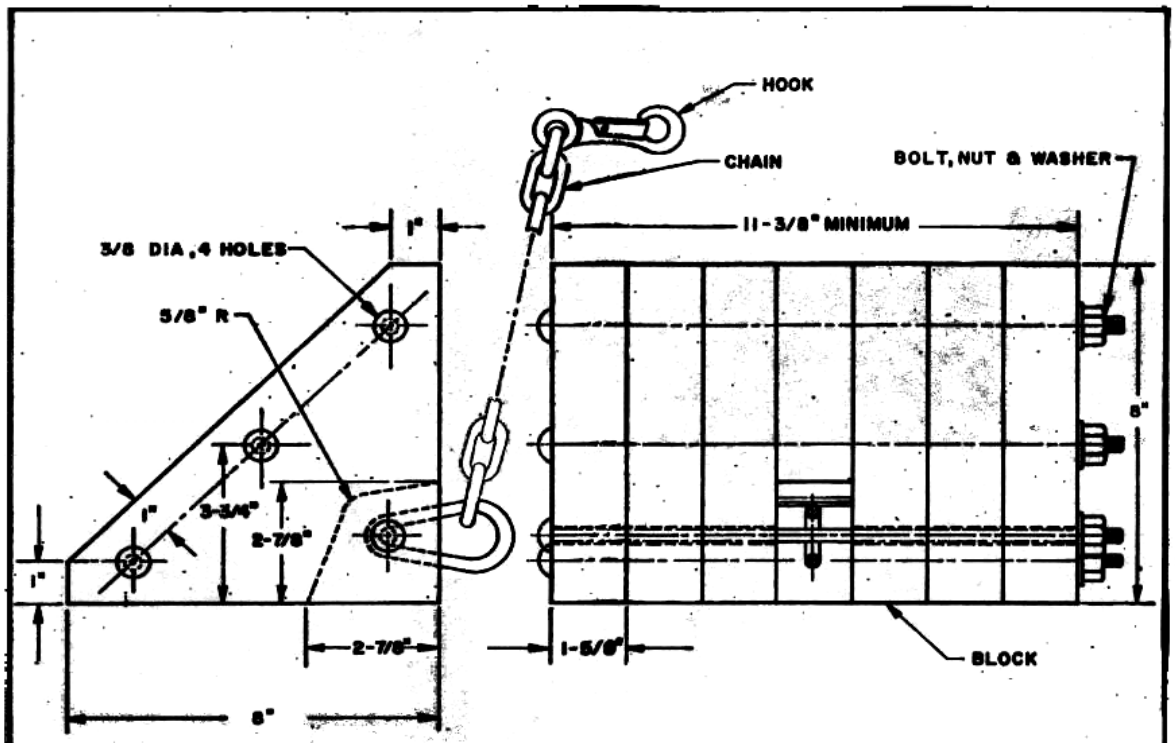
Navy – YD, MC

Preparing activity:

Army – ME

Project No. 3820-0069

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CHAIN DETAILS

CHAIN WELDED, CLOSE STRAIGHT LINK, GALV'D, 1/4" NOMINAL SIZE OF CHAIN BARS.

LENGTH OF 100 LINKS = 100 INCHES

MINIMUM TENSILE STRENGTH = 3,400 LBS.

WEIGHT / 100 FT. = 70 LBS.

MATERIAL NOTE

WOOD PARTS, FABRICATED, HARDWOOD.

FIGURE 1. CHOCK BLOCK ASSEMBLY

DX-603

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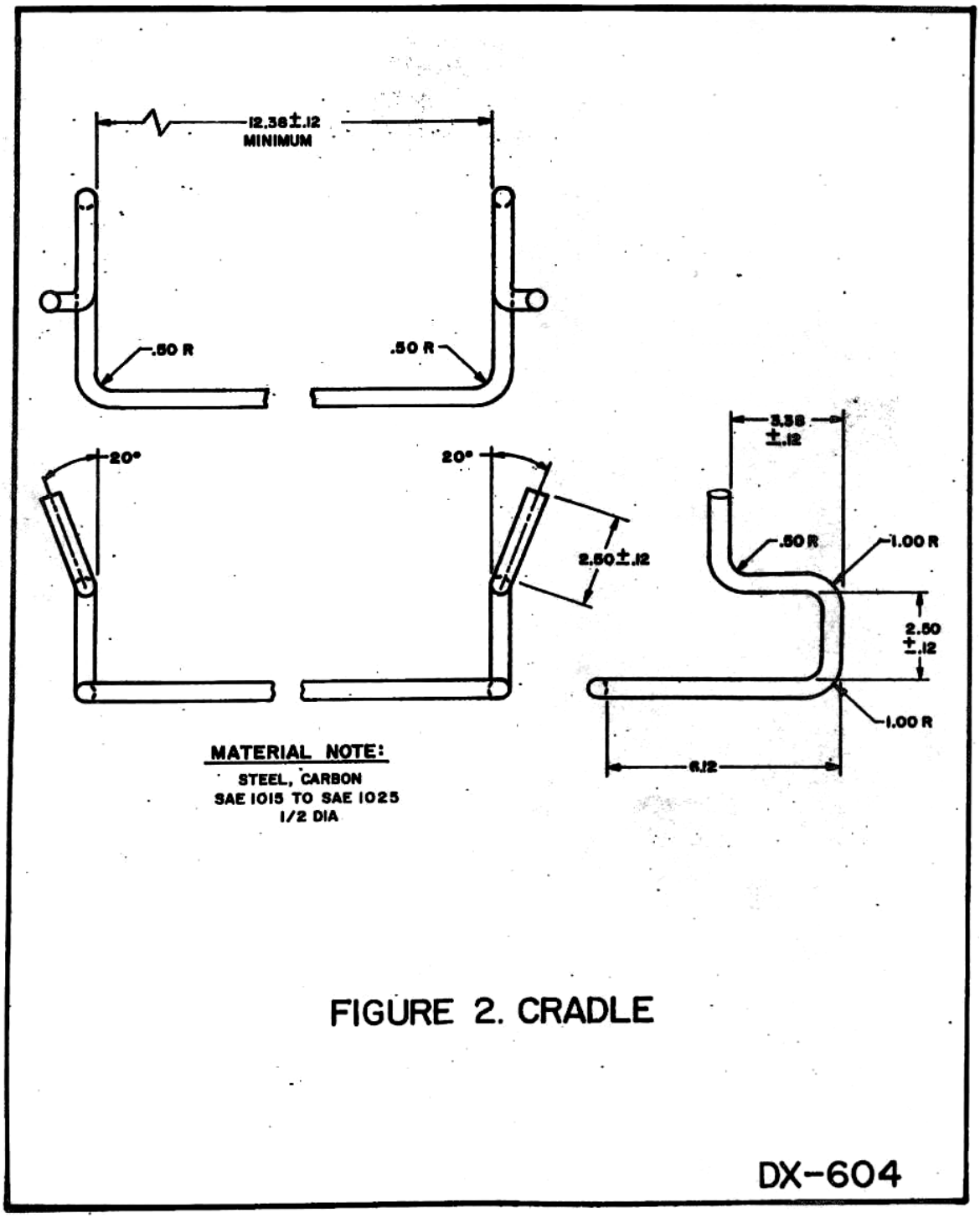
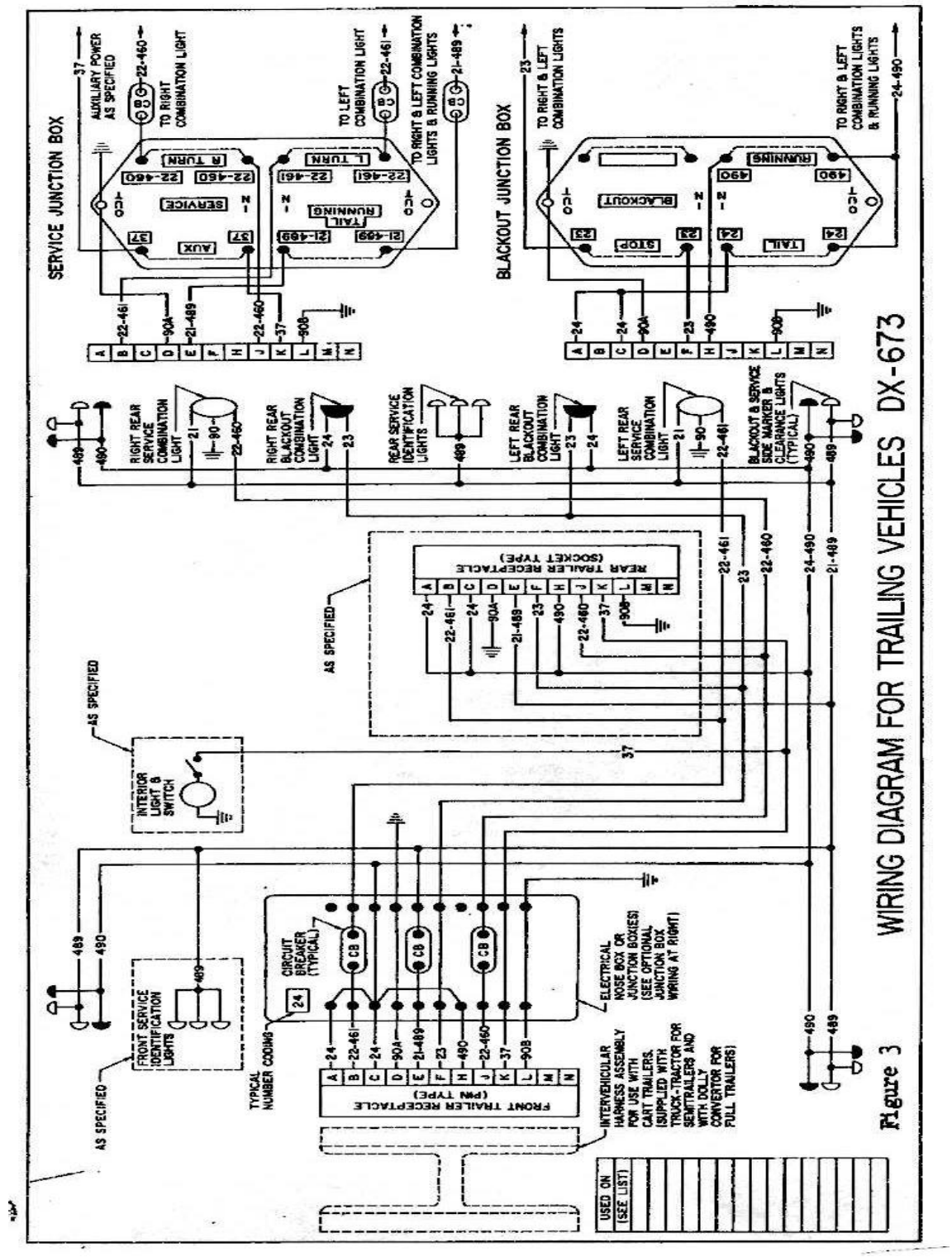
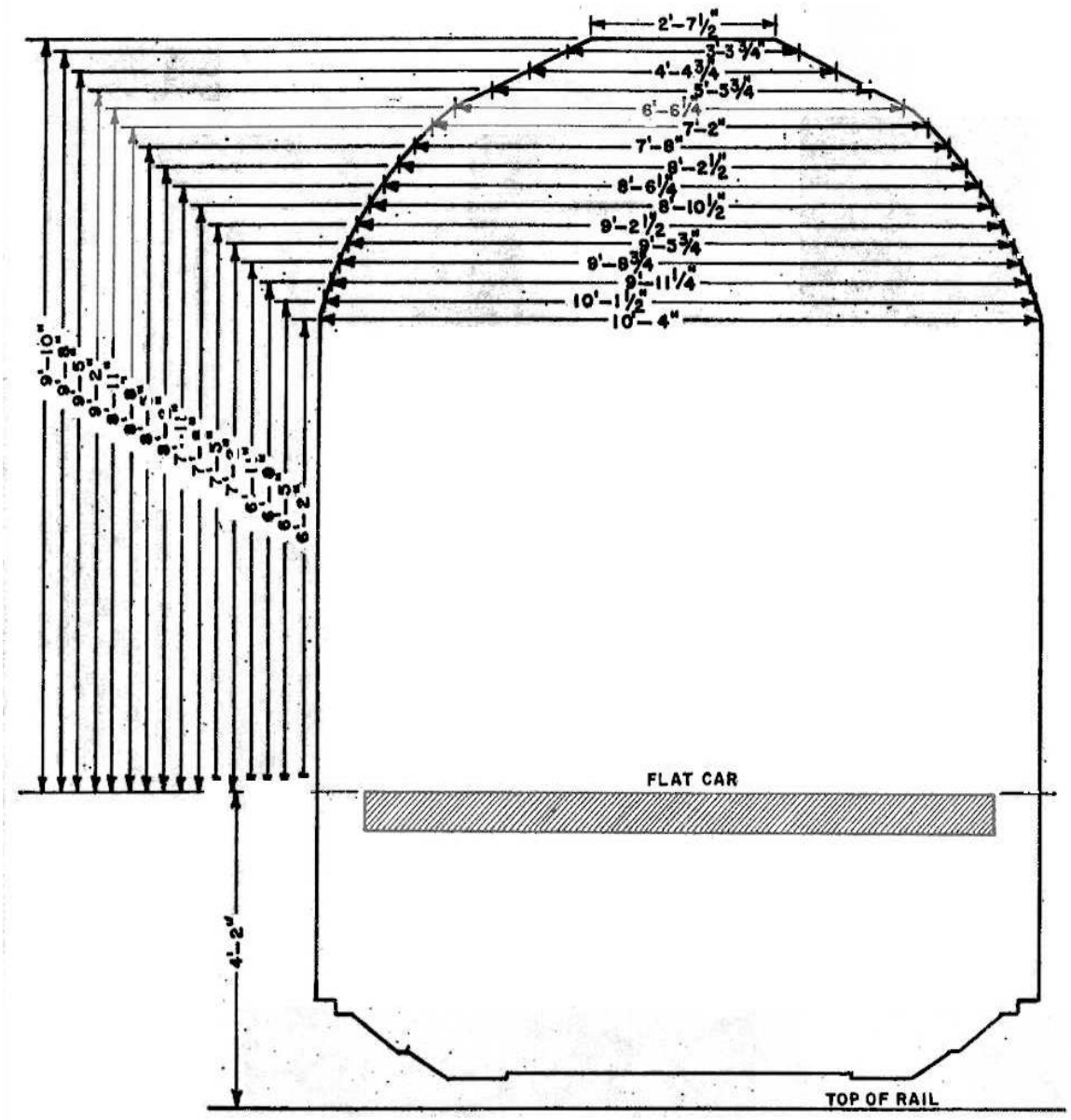


FIGURE 2. CRADLE

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ITEM LENGTH-32'-10"

FIGURE 4 BERNE INTERNATIONAL A13123-1
(CLEARANCE DIAGRAM)

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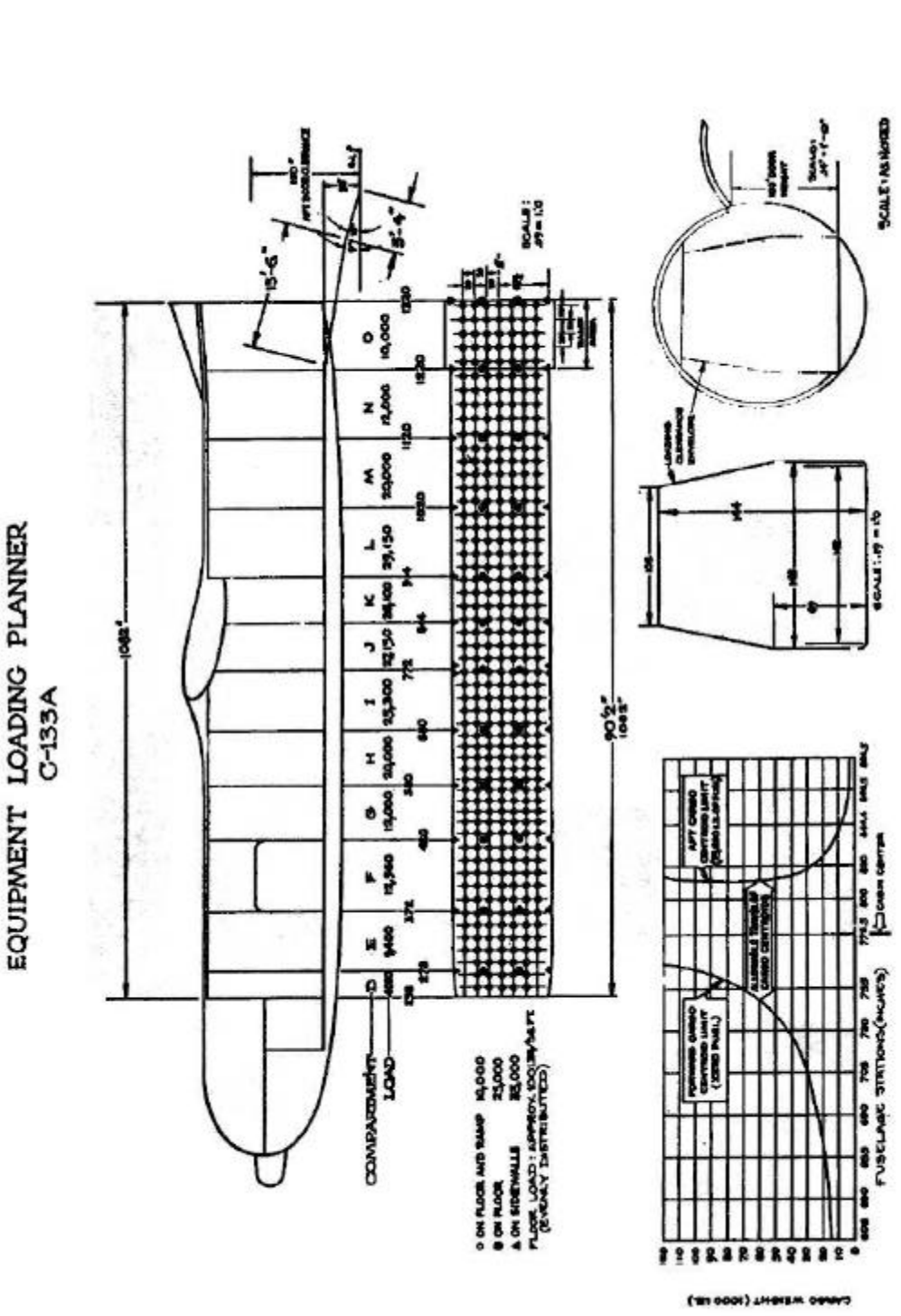


FIGURE 5.

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